Author: Mike Sigler at ~NMFS-ABL
Date: 12/03/96 04:18 AM
Priority: Normal
Receipt Requested
TO: Jeff Fujioka, Sandra Lowe at MAILHUB
Subject: FVOA logbook data

Jeff, Sandra,

I took a look at the FVOA logbook data. A spreadsheet which contains the cpue data is attached. Note that the RPN in fact is an RPW since catch is expressed in pounds, not numbers. Also the logbook RPW is expressed in pounds round weight and per 180 hook 1800 ft long skate (3.1 m spacing), whereas the survey RPW is kilograms round weight and per 45 hook 100 m long skate (2 m spacing).

1) There may be an error in NRC's computation of the logbook CPUE, because the survey catch rate usually is much greater than the logbook catch rate after standardizing the logbook CPUE to the survey standard.

2) The logbook RPWs by year-area-depth are area-weighted cpues (area X cpue), whereas the logbook RPWs by year-area mix effort and area weighting: First an area cpue is computed by effort-weighting each cpue, e.g.

cpue area 610 = (cpue201-300 X effort201-300 + cpue301-400 X effort301-400 + ...)/(effort201-300 + effort301-400+ ...)

then this area cpue is multiplied by area size to find the area RPW:

RPW area 610 = area size 610 X cpue area 610

This differs from computation of the survey RPW, where RPW by year-area is computed by summing RPW by year-area-depth across depth.

3) The trends of the survey and logbook catch rates are similar except for increased fishery catchability due to IFQ management.

Mike

12/2/96 Convert FVOA logbook data to NMFS survey standard

$$\text{FVOA cpue} = \frac{\text{catch in pounds}}{\text{180 hook, 3m spacing skate}}$$

FVOA std 180 hooks, 1800 ft skate

spacing = 3.1 m (10 ft)

$$\underset{3m\ sp.}{\text{catch/hook,}} = \frac{\text{cpue}}{180}$$

$$\underset{2m\ sp}{\text{catch/hook,}} = \underset{3m.\ sp.}{\text{catch/hook}} \times \underbrace{0.85}_{\text{spacing conversion}}$$

NMFS std 45 hooks, 100 m. skate

spacing = 2 m.

$$\text{FVOA cpue} = \frac{\text{catch in kg}}{\text{45 hook, 2 m spacing skate}} = \underset{2m.\ sp}{\text{catch/hook,}} \times \frac{1}{2.208} \times 45$$

Convert pounds to kilograms and standardize to 45 hook skate

FVOA RPW = FVOA cpue × area

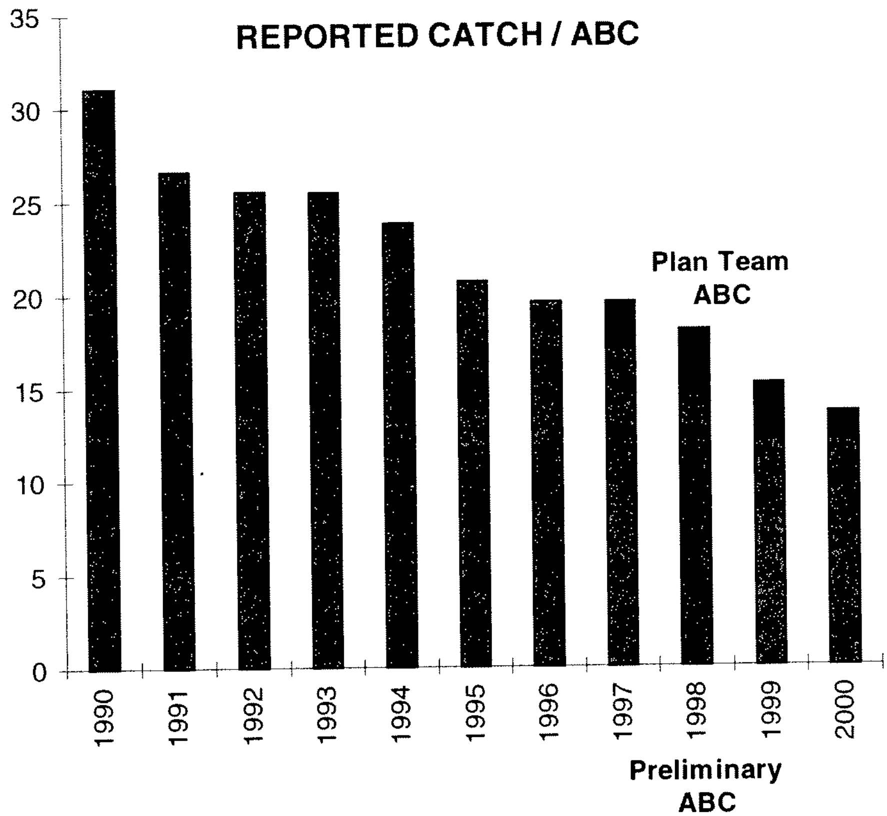
Conclusions

1) There may be an error in the computation of the logbook index, because the survey catch rate is usually much greater than the log book catch rate

2) The logbook indices by year-area-depth are computed correctly, but the values by year are not.

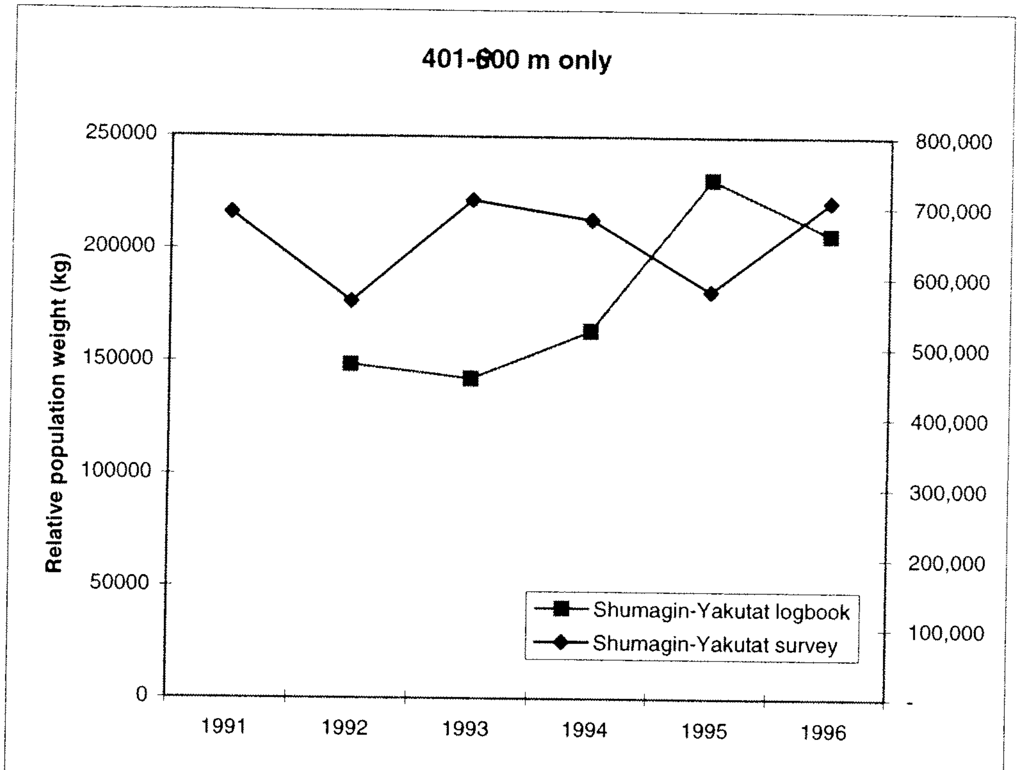
3) The trends of the survey and logbook catch rates are similar except for increased fishery catchability due to IFQ management.

C:\FISHYDAT\FVOA.XLS

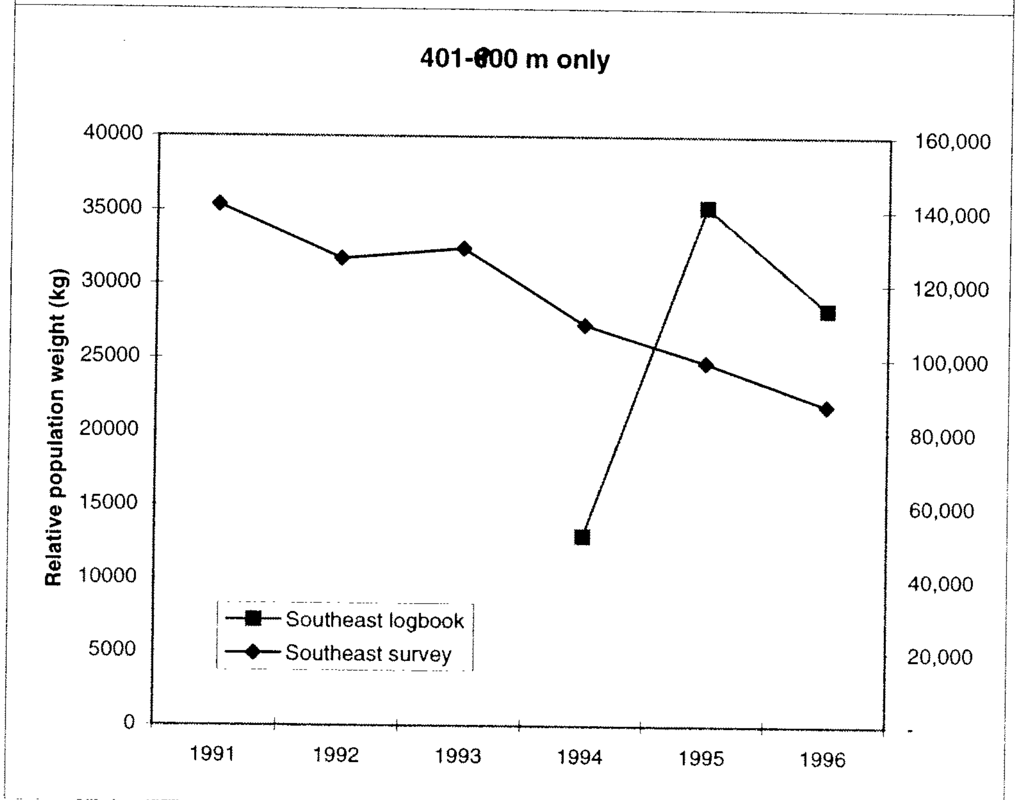


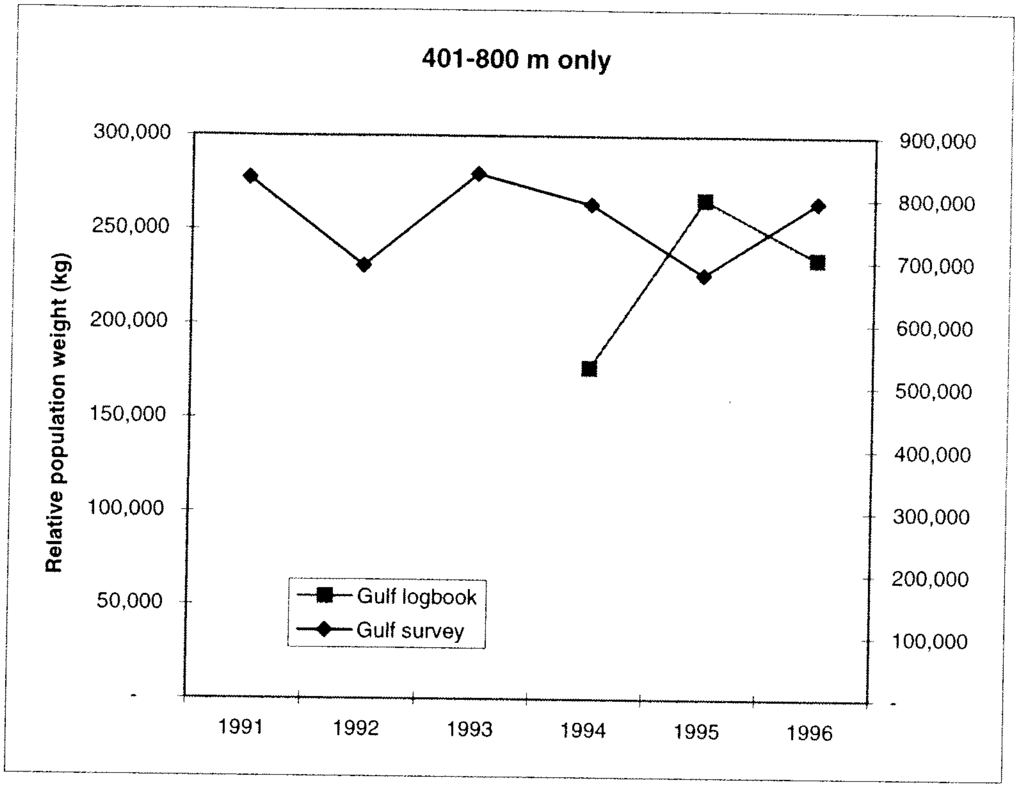
Probability that spawning biomass will fall below 95% of the lowest observed spawning biomass:

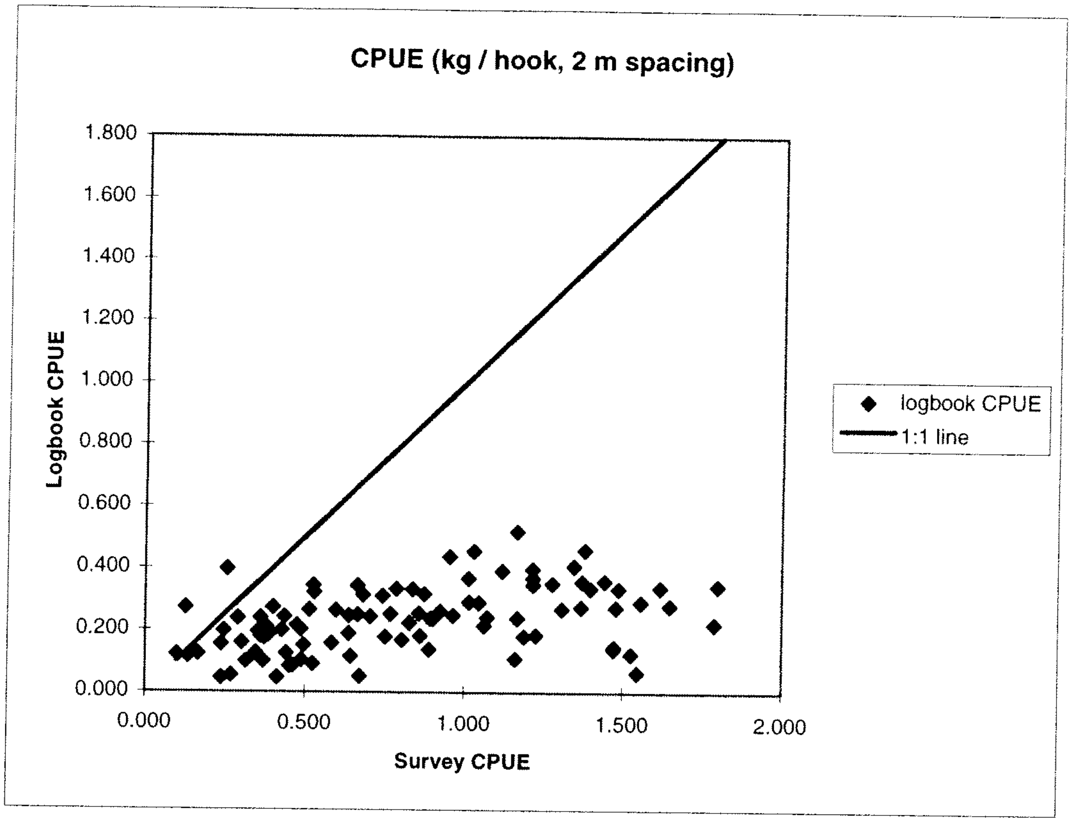
Year	Plan Team ABC	Preliminary ABC
1997	0.00	0.00
1998	0.00	0.00
1999	0.00	0.00
2000	0.65	0.21
2001	0.84	0.53



82% of effort within 401-800 m.




401-800 m only
Relative population weight (kg)
300,000
250,000
200,000
150,000
100,000
50,000
-
900,000
800,000
700,000
600,000
500,000
400,000
300,000
200,000
100,000
-
1991
1992
1993
1994
1995
1996
Gulf logbook
Gulf survey


CPUE (kg / hook, 2 m spacing)
Logbook CPUE
Survey CPUE
1.800
1.600
1.400
1.200
1.000
0.800
0.600
0.400
0.200
0.000
0.000
0.500
1.000
1.500
2.000
logbook CPUE
1:1 line

Effort (180 Hook Skates) by NMFS Depth Strata

GOA Federal Reporting Area 610

	Number of Skates (180 Hooks) Fished						
Depth (m)	1991	1992	1993	1994	1995	1996	All Years
201-300	0	536	0	0	0	0	536
301-400	0	80	0	0	29	0	109
401-600	360	1,259	92	29	492	1,152	3,384
601-800	157	2,016	137	219	1,311	1,573	5,412
801-1000	28	374	23	0	24	44	493
201-1000	546	4,265	251	247	1,856	2,769	9,934

GOA Federal Reporting Area 620

	Number of Skates (180 Hooks) Fished						
Depth (m)	1991	1992	1993	1994	1995	1996	All Years
201-300	0	0	0	0	0	0	0
301-400	0	0	43	0	0	0	43
401-600	0	496	440	600	311	74	1,921
601-800	0	910	847	794	799	1,990	5,341
801-1000	98	1,089	1,168	401	0	86	2,841
201-1000	98	2,495	2,499	1,795	1,109	2,151	10,145

GOA Federal Reporting Area 630

	Number of Skates (180 Hooks) Fished						
Depth (m)	1991	1992	1993	1994	1995	1996	All Years
201-300	204	857	356	678	667	314	3,076
301-400	183	74	167	7	0	62	494
401-600	47	612	1,396	276	613	1,013	3,957
601-800	141	591	1,634	782	3,829	2,746	9,724
801-1000	0	435	1,569	727	291	233	3,255
201-1000	575	2,570	5,122	2,470	5,401	4,367	20,506

Effort (180 Hook Skates) by NMFS Depth Strata

GOA Federal Reporting Area 640

	Number of Skates (180 Hooks) Fished						
Depth (m)	1991	1992	1993	1994	1995	1996	All Years
201-300	0	18	448	32	57	0	555
301-400	332	28	721	0	40	53	1,173
401-600	1,408	429	632	215	858	1,106	4,648
601-800	1,010	1,246	432	1,518	4,286	22,506	30,998
801-1000	0	803	299	582	131	109	1,924
201-1000	2,749	2,524	2,532	2,347	5,372	23,773	39,298

GOA Federal Reporting Area 650

	Number of Skates (180 Hooks) Fished						
Depth (m)	1991	1992	1993	1994	1995	1996	All Years
201-300	0	0	0	0	0	0	0
301-400	0	0	0	0	0	0	0
401-600	0	0	0	37	191	46	273
601-800	0	0	0	0	363	302	666
801-1000	0	0	0	0	37	0	37
201-1000	0	0	0	37	591	348	976

GOA Federal Reporting Areas 610-650

	Number of Skates (180 Hooks) Fished						
Depth (m)	1991	1992	1993	1994	1995	1996	All Years
201-300	204	1,411	805	710	724	314	4,167
301-400	515	183	931	7	69	115	1,819
401-600	1,815	2,796	2,560	1,155	2,465	3,392	14,182
601-800	1,308	4,763	3,050	3,313	10,588	29,117	52,140
801-1000	126	2,701	3,059	1,710	483	472	8,550
201-1000	3,968	11,853	10,405	6,895	14,330	33,409	80,859

Data from 10 vessels - 2,486 sets - 80,859 total skates fished

Skates were standardized to 180 hooks per skate

Raw CPUEs by NMFS Depth Strata

GOA Federal Reporting Area 610

	CPUE (Round Catch in Lbs per Skate)						
Depth (m)	1991	1992	1993	1994	1995	1996	All Years
201-300	0.00	46.02	0.00	0.00	0.00	0.00	46.02
301-400	0.00	72.12	0.00	0.00	86.21	0.00	75.86
401-600	88.36	59.31	54.01	84.14	100.56	123.08	90.17
601-800	93.34	56.31	42.21	58.23	113.22	124.15	90.61
801-1000	57.14	58.06	21.47	0.00	20.98	54.00	54.15
201-1000	88.19	56.36	44.62	61.22	108.26	122.58	86.09

0.48 #/hook

GOA Federal Reporting Area 620

	CPUE (Round Catch in Lbs per Skate)						
Depth (m)	1991	1992	1993	1994	1995	1996	All Years
201-300	0.00	0.00	0.00	0.00	0.00	0.00	0.00
301-400	0.00	0.00	150.00	0.00	0.00	0.00	150.00
401-600	0.00	116.84	118.41	155.06	160.39	135.67	136.90
601-800	0.00	121.78	135.01	111.70	160.69	155.20	140.65
801-1000	48.21	94.77	73.42	70.78	0.00	115.12	81.63
201-1000	48.21	109.02	103.55	117.05	160.61	152.92	123.45

GOA Federal Reporting Area 630

	CPUE (Round Catch in Lbs per Skate)						
Depth (m)	1991	1992	1993	1994	1995	1996	All Years
201-300	46.59	98.47	74.43	126.83	127.33	115.57	106.50
301-400	91.39	38.91	81.93	185.19	0.00	112.65	84.22
401-600	23.62	117.52	104.37	110.04	182.71	155.67	131.13
601-800	68.47	86.36	127.72	130.69	167.51	166.55	151.22
801-1000	0.00	78.29	100.38	84.96	110.95	145.89	98.18
201-1000	64.37	95.08	107.78	114.01	161.22	158.50	130.60

Raw CPUEs by NMFS Depth Strata

GOA Federal Reporting Area 640

	CPUE (Round Catch in Lbs per Skate)						
Depth (m)	1991	1992	1993	1994	1995	1996	All Years
201-300	0.00	83.33	111.27	25.00	92.83	0.00	103.50
301-400	144.75	64.42	113.67	0.00	39.83	111.79	118.69
401-600	156.29	125.56	58.03	162.33	203.71	241.87	169.49
601-800	135.58	157.91	160.79	103.56	214.17	29.89	69.39
801-1000	0.00	84.50	50.55	65.16	185.22	171.44	85.16
201-1000	147.29	127.49	99.94	98.35	209.21	40.58	83.96

GOA Federal Reporting Area 650

	CPUE (Round Catch in Lbs per Skate)						
Depth (m)	1991	1992	1993	1994	1995	1996	All Years
201-300	0.00	0.00	0.00	0.00	0.00	0.00	0.00
301-400	0.00	0.00	0.00	0.00	0.00	0.00	0.00
401-600	0.00	0.00	0.00	163.64	211.99	147.47	194.63
601-800	0.00	0.00	0.00	0.00	190.34	171.79	181.91
801-1000	0.00	0.00	0.00	0.00	128.18	0.00	128.18
201-1000	0.00	0.00	0.00	163.64	193.47	168.57	183.46

GOA Federal Reporting Areas 610-650

	CPUE (Round Catch in Lbs per Skate)						
Depth (m)	1991	1992	1993	1994	1995	1996	All Years
201-300	46.59	78.36	94.97	122.24	124.61	115.57	98.32
301-400	125.76	57.42	109.66	185.19	59.32	112.25	107.51
401-600	139.38	92.42	93.54	144.20	173.07	172.16	135.94
601-800	123.26	99.13	130.60	108.93	179.95	57.91	95.59
801-1000	50.20	83.98	84.62	74.90	128.01	137.50	87.34
201-1000	124.70	90.98	103.33	107.84	173.63	71.36	102.20

Data from 10 vessels - 2,486 sets - 80,859 total skates fished

Skates were standardized to 180 hooks per skate

NMFS Weighting Factors by Area and Depth Strata (km^2)

Weighted CPUEs - Relative Population Numbers

GOA Federal Reporting Area 610

Stratum Area (km2)	Depth (m)	RPNs CPUE (Round Catch in Lbs per Skate) X Stratum Area (km^2) 1991	1992	1993	1994	1995	1996	All Years
2,737	201-300	0	125,969	0	0	0	0	125,969
1,264	301-400	0	91,165	0	0	108,966	0	95,883
2,269	401-600	200,485	134,576	122,551	190,914	228,168	279,259	204,604
1,629	601-800	152,050	91,727	68,758	94,862	184,436	202,239	147,605
1,248 ~~1,264~~	801-1000	73,371	74,550	27,566	0	26,937	69,336	69,529
9,147	201-1000	806,698	515,484	408,173	560,004	990,217	1,121,211	787,427

GOA Federal Reporting Area 620

Stratum Area (km2)	Depth (m)	RPNs CPUE (Round Catch in Lbs per Skate) X Stratum Area (km^2) 1991	1992	1993	1994	1995	1996	All Years
1,533	201-300	0	0	0	0	0	0	0
817	301-400	0	0	122,550	0	0	0	122,550
1,766	401-600	0	206,338	209,112	273,836	283,243	239,596	241,771
1,955	601-800	0	238,087	263,945	218,375	314,153	303,409	274,974
2,012	801-1000	96,989	190,684	147,721	142,418	0	231,614	164,232
8,083	201-1000	389,642	881,174	836,956	946,140	1,298,181	1,236,036	997,869

GOA Federal Reporting Area 630

Stratum Area (km2)	Depth (m)	RPNs CPUE (Round Catch in Lbs per Skate) X Stratum Area (km^2) 1991	1992	1993	1994	1995	1996	All Years
1,626	201-300	75,763	160,111	121,018	206,231	207,032	187,914	173,161
1,480	301-400	135,259	57,581	121,256	274,078	0	166,715	124,642
2,255	401-600	53,265	265,015	235,360	248,149	412,015	351,040	295,694
1,923	601-800	131,661	166,065	245,601	251,308	322,125	320,276	290,792
2,296	801-1000	0	179,753	230,466	195,058	254,750	334,952	225,430
9,580	201-1000	616,654	910,899	1,032,561	1,092,204	1,544,519	1,518,423	1,251,153

NMFS Weighting Factors by Area and Depth Strata (km²)

Weighted CPUEs - Relative Population Numbers

GOA Federal Reporting Area 640

Stratum Area (km2)	Depth (m)	RPNs CPUE (Round Catch in Lbs per Skate) X Stratum Area (km²) 1991	1992	1993	1994	1995	1996	All Years
1,494	201-300	0	124,500	166,243	37,350	138,682	0	154,636
1,494	301-400	216,260	96,242	169,816	0	59,499	167,013	177,319
1,666	401-600	260,379	209,186	96,670	270,434	339,384	402,951	282,375
1,470	601-800	199,308	232,128	236,356	152,236	314,835	43,944	102,010
1,489	801-1000	0	125,819	75,272	97,024	275,790	255,277	126,798
7,613	201-1000	1,121,335	970,603	760,869	748,751	1,592,711	308,969	639,172

GOA Federal Reporting Area 650

Stratum Area (km2)	Depth (m)	RPNs CPUE (Round Catch in Lbs per Skate) X Stratum Area (km²) 1991	1992	1993	1994	1995	1996	All Years
891	201-300	0	0	0	0	0	0	0
891	301-400	0	0	0	0	0	0	0
822	401-600	0	0	0	134,509	174,258	121,220	159,984
1,006	601-800	0	0	0	0	191,482	172,825	183,006
1,165	801-1000	0	0	0	0	149,332	0	149,332
4,774	201-1000	0	0	0	781,200	923,634	804,777	875,830

GOA Federal Reporting Areas 610-650

Stratum Area (km2)	Depth (m)	RPNs CPUE (Round Catch in Lbs per Skate) X Stratum Area (km²) 1991	1992	1993	1994	1995	1996	All Years
8,281	201-300	385,853	648,927	786,412	1,012,309	1,031,896	957,019	814,225
5,946	301-400	747,760	341,441	652,028	1,101,128	352,710	667,449	639,237
8,778	401-600	1,223,515	811,247	821,084	1,265,788	1,519,202	1,511,247	1,193,239
7,983	601-800	983,999	791,365	1,042,557	869,570	1,436,508	462,296	763,106
8,210	801-1000	412,135	689,468	694,746	614,902	1,050,924	1,128,897	717,058
39,197	201-1000	4,887,916	3,566,214	4,050,328	4,227,184	6,805,954	2,797,127	4,006,087
		11.4	9.5	9.5	12.18	19		

0.57#/hook

Not an area-weighted cpue

Data from 10 vessels - 2,486 sets - 80,859 total skates fished

Skates were standardized to 180 hooks per skate

Sum of Num_Skates		NMFSDepCat					
Year	Vessel	201-300	301-400	401-600	601-800	801-1000	Grand Total
91	Masonic	0	0	236	54	28	318
	Quest	161	23	695	769	0	1648
	Seymour	0	0	0	0	90	90
	Vansee	47	444	830	475	0	1796
91 Total		208	467	1761	1298	118	3852
92	Allstar	0	0	466	878	812	2156
	Masonic	0	10	224	652	144	1030
	Polaris	0	42	512	1198	514	2266
	Quest	544	30	483	598	519	2174
	Seymour	43	65	382	533	415	1438
	Tordenskjold	738	25	30	280	271	1344
	Vansee	0	0	607	500	0	1107
92 Total		1325	172	2704	4639	2675	11515
93	Allstar	0	0	150	639	756	1545
	Augustine	495	592	45	80	0	1212
	Lorelei II	70	95	214	428	537	1344
	Polaris	0	0	256	558	425	1239
	Quest	212	96	604	86	170	1168
	Seymour	0	40	364	367	460	1231
	Tordenskjold	0	47	176	494	500	1217
	Vansee	0	18	627	297	141	1083
93 Total		777	888	2436	2949	2989	10039
94	Allstar	0	0	190	325	170	685
	Augustine	665	12	216	116	0	1009
	Lorelei II	263	0	35	277	50	625
	Masonic	0	0	155	502	324	981
	Polaris	0	0	130	395	132	657
	Quest	32	0	191	835	158	1216
	Seymour	0	0	255	433	226	914
	Tordenskjold	0	0	0	136	484	620
	Vansee	0	0	60	257	90	407
94 Total		960	12	1232	3276	1634	7114
95	Allstar	0	0	450	822	60	1332
	Augustine	697	25	677	933	0	2332
	Lorelei II	27	44	594	1334	65	2064
	Masonic	0	0	165	858	0	1023
	Polaris	0	0	46	674	192	912
	Quest	0	0	90	2624	86	2800
	Seymour	0	0	298	954	22	1274
	Tordenskjold	0	0	100	812	50	962
	Vansee	0	0	44	1426	0	1470
95 Total		724	69	2464	10437	475	14169
96	Allstar	0	0	675	487	0	1162
	Augustine	272	100	990	110	0	1472
	Grant	0	0	601	1201	0	1802
	Lorelei II	0	12	363	20642	75	21092
	Masonic	0	0	236	1640	60	1936
	Polaris	0	0	20	402	186	608
	Quest	44	0	363	1460	46	1913
	Seymour	0	0	116	1051	24	1191
	Tordenskjold	0	0	25	674	0	699
	Vansee	0	0	0	1142	64	1206
96 Total		316	112	3389	28809	455	33081
Grand Total		4310	1720	13986	51408	8346	79770

DATE:	November 19, 1996
TO:	FVOA
FROM:	DLA

DRAFT

In accordance with our agreement Natural Resources Consultants, Inc. has:

1. Evaluated 10 boats from your fleet and developed CPUE by area and depth intervals throughout the Gulf. The data involved 2,519 sets.

2. Reviewed the current NMFS survey procedure and sampling methodology.

3. Reviewed the DD (Delayed Differences) and AB (Age Based) models in terms of assumptions, model structure, spreadsheets and mathematical procedures.

4. Recommended a set of interaction with the NMFS/AFS.

Survey Methodology

The annual NMFS survey involves deployment of 160 skates. A standard skate is 100 m long with 45 hooks. The longline is set normal to the depth (depth contours) in a manner to cover a depth range from 101 m - 1000 m. As the distance over this depth range varies between stations all depth ranges cannot be covered at some stations (when distance exceeds 8.6 nautical miles). Fishing depth is recorded for each 5 skates of gear and catch in numbers of major groundfish species are recorded for each skate of gear.

Catch data are allocated to 8 depths strata as follows: (1) 0 m - 100 m, (2) 101 m - 200 m, (3) 201 m - 300 m, (4) 301 m - 400 m, (5) 401 m - 600 m, (6) 601 m - 800 m, (7) 801 m - 1000 m, (8) 1001 m - 1200 m. These strata were later combined over some of the depth intervals. In analysis of the data CPUE in numbers of fish caught per skate are determined for each region, station and depth zone. This gives a mean catch rate for each stations depth strata. The values are subsequently multiplied (weighted) by the square kilometer (km^{2}) involved in each area depth strata. The "relative population numbers" (RPN), are averaged across strata to complete a regional strata RPN. These regional values are then added to obtain a regional slope (201 m - 1000 m) RPN.

In the old Delayed Difference (DD) model these values (RPN) were converted to relative catch per weight (RPW) based on size and weight, information and subsequently calibrated to swept area trawl data to obtain an absolute biomass value for the Gulf and other regions. The survey constitutes one of the few options for sablefish assessment, but the calibration factor could be misleading. The line survey method is designed to yield a relative population index, which is assumed to follow the trend in absolute exploitable population size. The methods used to set ABC and TAC's up until this year was based on the DD model, which estimated absolute population size using a comparative trawl/longline calibration. The underlying assumptions in the DD model and the trawl/longline calibrations over time require a good deal of faith as compared with the newly adopted age based (AB) model.

The AB model, in our view, provides a more reliable technique to track sablefish population and established populations yields. The AB model relies on the same RPN values but uses the trends in these numbers and does not need an estimate of absolute numbers or population weight. It is depended on the <u>trend</u> being accurate.

Both methods rely on the estimate of the total fish in the sea available for harvest and the proportion available for harvest to establish ABC. A comparison of the models and data show that the population estimates resulting from the two methods differ (about 16%) <u>with the traditional DD model yielding a population estimate of 193,000 mt and the new AB model about 162,000 mt.</u> This difference by itself would seem to require a reduction in ABC of 16% over the previous year.

A compensating difference, however, occurs when we look at the exploitation proportion associated with the 35% of unexploited biomass rule. → due to different growth relationship, selectivity
The exploitation estimated by the AB method (11.4) is about 9% larger than that for the DD method (10.6). A similar difference occurs in the proposed exploitation portions associated with the 40% of under exploited biomass rule. Putting the difference together and the exploitation proportions results in as AB model estimate of ABC of 18,500 (under the 35% rule) and 19,600 mt using the old DD model (at the 35% rule).

The 1997 ABC estimate, not adjusting for new data that will be supplied by the 1996 survey leads to an estimated ABC (based on the AB model) of 13,3000 mt significantly lower than expected from the population decline. In conclusion it is clear from the above observations that the major reduction required in 1997 will result from the implementation of the (40%) rule. A very conservative management policy.

The preliminary estimate of the 1997 ABC is substantially smaller than the 1996 ABC. This is partly because the biomass is predicted at a lower level for 1997 by the new AB model but largely because of the new management rules for settling ABC which are more precautionary than those used to set the 1995 ABC. It should be noted however that these preliminary numbers could be significantly;y changed based on the 1996 survey data and exploration of attendant information.

About the Model and Calculations

1. Our examination of the calculations made to estimate the survey biomass trend and the new AB assessments method did not reveal any errors, rather we conclude they were made in a professional manner, albeit in rather a preliminary fashion.

2. The model used appears logically coherent and in fact one of the few age based approaches possible given the sparse nature of available data from this stock.

3. The spread sheet implementations, while clearly provisional, have been constructed in a professional manner and as far as we can see free from programming errors.

4. NRC considered whether a more detailed and better known age based model (such as ad hock tuned VPA) might be adopted to the sablefish stock, but found that the data would not adequately support this approach.

Thus we conclude the science is well done and the major reduction in fishable yield reflects the more conservative policy adopted by management. The situation of the sablefish seems to be in control but the stock decline is of course a concern. In our view at the present it requires a careful rather than excessively conservation management approach. If the declining recruitment continues to be low and the stock continues to decline then the future may require a more stringent measures. The new management rules applied with discretion would seem adequate to the task.

Some Questions and Recommendations to the Planning Team

1. As far as we can ascertain the new rules for precautionary management are set out in six descending tiers. These tiers are arranged in descending order in correspondence to management information available. The preliminary 1997 ABC seems to have been set under the third tier. If we are correct, then (a) if the stock is above B40%, then FOFL is set equal to F30% and FABC is set at or less than 40%. As the current biomass is less than 40% but greater than

the arbitrary limit set of 5% of B40%, FOFL and FABC are set by a rather conservative formula e.g. F30% x (B/B40% - 0.05/.95 and F40% x B/B40% - .05)/-95. The effort of these two formulas are to progressively reduce FOFL and FABC. Question. Is this the appropriate tier based on the information currently available for management? What new data source or analysis would be required to improve the tier level?

2. We note that the AB model has followed the relative biomass trajectory using RPN. Why not RPW? Reworking the age based assessment with different input assumptions made it possible to reconcile the model predictions more closely with the RPW longline survey index of total stock abundance. Giving less weight to the length distribution data and turning the results to the RPW rather than the RPN increased the compatibility of the new method to the DD model and resulted in modest but significant increases in the ABC and OFL catch levels for 1996. Question. Why are the RPN used to follow population trend rather than RPW?

3. The model rest on several assumptions that need some exploration e.g. (1) the age structure information gathered from the long line survey is representative of that in the commercial fishery, (2) the commercial catch is accurately reported and known, (3) males and females constitute equal portions of the population and (4) age specific mortality is the same over the age ranges used in the model.

We believe priority should be given to sampling of the commercial landings to first check assumptions number one.

Results of NRC analysis of log data are provided in Tables 1-3. The information has been summarized by areas and depth strata in a manner similar to that done for the NMFS longline survey. Data are provided for GOA areas 610, 620, 630,640, 650 and 610 to 650 combined. Table 1 sums raw effort by area and depth while Table 2 gives the calculated catch per unit effort for the same areas and depth cells. Table 3 provided the cell CPUE data weighted by the area square kilometers (km) for each area and depth zone (RPN) and areas 610-650, all depth zones and the 201-1000 depth zone summarized. Finally, Table 4 provides a listing of vessels participating and numbers of skates set.

This data tracked quite well with the NMFS sablefish index values of RPN from 1991 through 1993, but not in the past two years. The commercial log book data show a sharp rise in 1995 and a significant drop in 1996 years in which the ITQ system was introduced. However, ignoring 1995, the first year of the ITQ system, 1996 is down about 31% over 1994 while the data from NMFS is down about 20%.

It is interesting to note the average RPN values for the commercial data are an order of magnitude higher than that shown for the NMFS longline survey. This in part reflects the large size of the standard skate (1800 ft 2 vs. 3.1 m
versus about 328 ft. ~~416~~ ft) and 45 hooks/skate versus 180 hooks/skate, but also the hook spacing
capacity of the fleet to concentrate on the fishing areas yielding higher catch rates. Catch rates even when adjusted for number of hooks per skate are 2 - 3 times greater than the longline survey. However, some of this

difference may relate to soaking time. The results are what might be expected in that the longline survey is derived to sample the population over the Gulf area while commercial fishermen concentrate on areas where returns per effort are high.

The following data are attached:

1. Results of commercial fishery log information evaluated in terms of CPUE and RPN.

2. Pope's notes and comments on review.

3. Pope's original analysis of AB and DD models along with comments and suggestions.

4. List of vessels and effort provided to NRC.

DLA

Ministry of Agriculture, Fisheries & Food

Directorate of Fisheries Research

FACSIMILE TRANSMISSION

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Number of pages including this one: .... Date : ....

MESSAGE:

Dear Lee

I am sending a rough draft for you comments.

As you can see it still needs some amendments which I will get on with.

Report on Changes in the Sablefish Assessment.

1 *Introduction*

Why do Acceptable Biological Catches(ABC's) Change?

All estimates of allowable catches rely on two components. These are:-

1. The total weight of fish in the sea available for capture by fishermen (We call this the exploitable biomass)
2. The proportion that management proposes, that the allowable catch should form of the exploitable biomass (We will call this the proposed exploitation proportion).

Allowable catches will thus change if either there are more or less exploitable fish in the sea, and if the management agency decides to allow a greater or less proportion of these to be exploited. They may also change, if the methods or data used by fisheries scientists, to estimate the exploitable biomass and the proposed exploitation proportion change.

The second of the two effects can be seen in the changes in the Preliminary Allowable Biological Catches (ABC's) estimated by NMFS scientists in the Preliminary 1997 Sablefish assessment report of Fujioka *et al* (1996). This is because the management target is set to change in 1997 from the 35% of unexploited population rule to the 40% of unexploited population rules, for setting proposed exploitation proportions(These rules are explained in the box). The preliminary ABC for 1997 has also changed a result of changing the scientific assessment methodology from the delay difference (DD) method, used in past years, to an age based (AB) method. However, the first effect noted above, change of exploitable biomass, does not yet act on the 1997 ABC. This is because the preliminary 1997 ABC is based upon the 1996 exploitable biomass estimates and not the 1997 exploitable biomass estimates. This is presumably because the results of the 1996 long-line survey results and the commercial catch data for 1996 are not yet available. Therefore, the 1997 exploitable biomass, and hence the 1997 ABC, may change when these new data become available. We may thus ask why the preliminary 1997 ABC has changed and we could also ask how this might change further when the final estimates are available? We might also ask what were the relative importance of the three factors; exploitable biomass change; management target changes for exploitation proportion; changes in scientific methodology; to the changes in the 1997 ABC?

1.1 What was the relative *importance of the factors that caused* changes *in the* Preliminary 1997 ABC estimate.

Figure 1 shows the 1996 exploitable biomass, the exploitation proportions that would satisfy the 35% of unexploited population rule and the 40% of unexploited population rule, and the resulting ABC's associated with these two rules. These results are shown both for the assessment based upon the delay difference analysis (DD) used in previous years and the age based analysis (AB) used in the current assessment. The figure shows that the 1996 estimate of exploitable biomass varied between the two assessment methods. That for the AB was smaller at 162,000 mt than the DD at 193,000 mt. This difference, by itself, would tend to give allowable catches that were 16% lower if the AB method was used compared to the DD method.

A compensating difference, however, occurs when we look at the proposed exploitation proportions associated with the 35% of unexploited population rule. The exploitation proportion estimated by the AB method (11.4%) is 8% larger than that for the DD method (10.6%). A similar 8% difference occurs in the proposed exploitation proportions associated with the 40% of unexploited population rule; 8.3% for AB and 7.6% for DD. Putting together the changes in the exploitable biomass and the changes in the proposed exploitation proportions, result in ABC's based upon the AB method which are 5% lower for the 35% rule (18500 mt.) than the equivalent result for DD (19600 mt.), and 5% lower for the 40% rule (AB=13300 mt. and DD=14000 mt.).

In these calculations the 35% rule result for the DD method (19600 mt) is effectively (apart from minor rounding errors) the 1996 ABC. In the preliminary calculations of the 1997 ABC, as calculated by the DD method, this has dropped to 14000 mt (i.e. by 28%). This is due to the change of the management target for the exploitation proportion, from the 35% rule to the 40% rule. It drops a further 4% because the Preliminary Assessment, based upon the 40% rule, was estimated by the AB method rather than by the DD method. The perceived change from the 1996 ABC to the predicted 1997 ABC is thus mostly due to the change in management objective from the 35% rule to the 40% rule (28%) but is also due to the change in method (4%). It is not due to a change in biomass as such because both the 1996 (35% rule) ABC and the preliminary 1997 (40% rule) ABC are both based upon estimates of the 1996 exploitable biomass. However, part of the change results from the fact that the perceived size of the 1996 exploitable biomass estimate changes depending on the method and data used to calculate it.

1-2 **What might be the relative importance of the factors that may caused changes in the final 1997 ABC estimate.**

Ultimately we presume the 1997 ABC will be based upon a 1997 estimate of exploitable biomass (calculated when the 1996 survey results and provisional 1996 catch estimates are available). This can be expected to change from the 1996 level of exploitable biomass. Given recent trends this might be expected to be downward and this could result in a further reduction of the 1997 ABC. Figure 2 shows estimates of the 1997 ABC that are based on biomass projections contained in spreadsheets supplied by NMFS scientists. Readers should note that these are TENTATIVE since they DO NOT include the 1996 survey results and provisional 1996 catch estimates in the assessment and they are included for illustrative purposes only. However the industry should note that the final estimate could possibly be lower than the provisional 13300 mt. estimate.

In Figure 2 it will be noted that the estimate of the 1997 exploitable biomass estimated by DD has fallen by 3% to 186,000 mt. A fall in biomass when the biomass is lower than the target level (see box on 35% and 40% rules) leads to a greater reduction in ABC because the exploitation proportion is then adjusted downward under both the 35% and the 40% rules. Thus the DD estimate of the 1997 ABC under the 35% rule (19,000 mt.) is 3% less than the ABC in 1996. If the 40% rule is adopted (as is planned for 1997) the ABC estimated by the DD method (13,000 mt.) is 29% (i.e. a further 26% lower) than the 1996 ABC. If the AB method of assessment is used then the ABC calculated under the 40% rule (11,000 mt.) is 43% lower (a further 14%). Thus the 43% change in this TENTATIVE 1997 ABC from the 1996 level might be approximately ascribed as being 3% due to biomass change, 26% due to the change in management policy and 14% due to the change in assessment methodology.

1.3 summary and aims

It is clear from the above calculations that the main changes in the preliminary estimate of the 1997 sablefish ABC are due to the proposed change in management policy from the 35% rule to the 40% rule. However, methodology change has had some effect on the preliminary estimate and might possibly (this can only be judged in the light of the final 1996 assessment) be more important in the final assessment of the 1997 sablefish ABC. Changes in the biomass between 1996 and 1997 may also affect the final 1997 sablefish ABC. Current indications are that this may be a fairly minor effect (at least as projected by the DD method) but the perception of this might change in the light of the results of the 1996 long-line survey.

The purpose of this investigation is thus to examine:-

Need to get changes [illegible]?

1. Are biomass changes being tracked accurately?
2. Are the changes in management policy a). Justified? b)Correctly calculated?
3. Is the new methodology proposed for estimating the 1997 ABC a). Sound? b). has it been used accurately?

Ref

Fujioka *et al* (1996).

Fig. 1 Development of the preliminary 1997 ABC
(Based up the NMFS for prediction using the 1996 exploitable biomass)

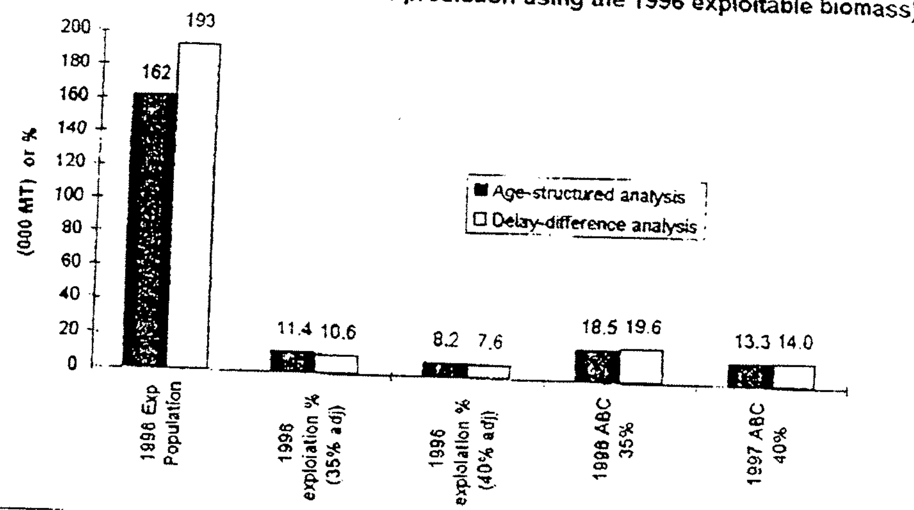
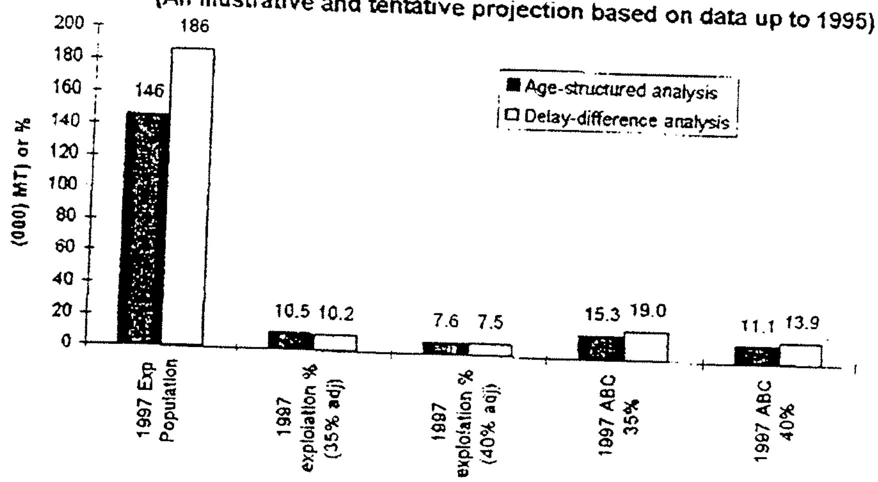


Fig 2 Development of the 1997 ABC
(An illustrative and tentative projection based on data up to 1995)



2 Are biomass changes being tracked accurately?

2.1 Introduction

It was noted in the introduction that the level of exploitable biomass is one of two main elements in the calculation of an ABC. It was also noted that it was not directly implicated in the reduced level of the preliminary 1997 ABC relative to the 1996 ABC but that change in it might well influence the level of the final 1997 ABC.

As well as being a measure of what weight of fish are available to catch, the level of the exploitable biomass also determines how the 35% and 40% rules will operate to reduce the proposed exploitation proportion, for the Delay Difference (DD) model calculations of ABC. The closely related spawning biomass (the weight of fish available to spawn) has a similar role when the Age Based Assessment method (AB) is used to calculate the ABC. Thus both the level and the trend of the exploitable biomass has a large potentially effect on the level of ABC's and requires tracking accurately. In the sablefish assessments, by both the DD and the AB models, abundance indices from long-line surveys are vital components for estimating exploitable biomass.

The Delay Difference model (DD) uses estimates of the total sablefish biomass (Relative Population Weight (RPW)) obtained from long-line surveys from the Eastern Bering sea, the Aleutian Islands Region and the Gulf of Alaska. These are calibrated to absolute biomass using comparisons made between long-line survey results and groundfish trawl survey results conducted in the ??? in 19?? Ref ???. The model uses these results as givens and tracks them exactly (CHECK THIS POINT). It is thus dependent upon the surveys indices tracking the trend in biomass and estimating the absolute biomass, if the DD method is to give reliable estimates of fishing intensity and ABC's.

The Age Based Assessment (AB) method uses estimates of the relative population numbers (RPN) obtained from long-line surveys from the Eastern Bering sea, the Aleutian Islands Region and the Gulf of Alaska.(CHECK ITS FROM ALL AREAS). This is one of the data sets to which the model is calibrated. It uses the trend in these results but not an estimate of absolute numbers. It is thus dependent only on the trend being accurate.

The available surveys do not span the entire period of the analyses and therefore it is necessary for them to be inter-calibrated. Additionally, in the case of the earlier years in the Bering sea and the Aleutian Island region, missing values have to be made up. Moreover, in the case of the RPW series, it is necessary to calibrate indices to absolute biomass.

Since the trends of survey indices influence the estimates of exploitable biomass made by both models it is necessary to check whether the inter-calibration of the separate series has been carried out in a reasonable fashion. Since the absolute level of the RPW series influences the DD model it is also necessary to check how this calibration was made. Finally, it was noted in the introduction that the exploitable biomass for 1996 estimated by the DD method was larger than that estimated by the AB method. Since the 1996 DD method's estimate of exploitable biomass is essentially a one year projection from the RWP series, and since this is calibrated to trawl survey swept area results (assuming that all sablefish in the path of the trawl were caught), it might have been expected to underestimate exploitable biomass compared to the AB estimates (because not all fish in the trawl path might have been caught). The fact that this was not the case ~~thus~~ seems curious and looks like a loose end that needs examination. The following subsections thus examine the inter-calibration of the survey series in detail and tug at the loose end.

2.2 Inter-calibration of the RWP index.

The inter-calibration of the RWP index is described in Fujioka (1995). Detailed calculations were available to us from spreadsheet DELAYDIF.XLS page rpwcatch, (~~kindly made available to us~~ by NMFS scientists). We have conducted an audit of the calculations made on the sheet and these are described below (Lee we probably should move this to a technical appendix). Indices are calibrated separately for the Eastern Bering Sea, The Aleutian Island area and the Gulf of Alaska and then combined. The last area is by far the most important and we deal with its calibration first for two separate time periods.

This was what was

2.2.1 Gulf of Alaska RWP indices 1990 to 1995. (page rpwcatch ref. E19-E24)

2.2.1.1

These are based upon the Domestic survey series and constructed as follows. The spreadsheet combines the GOA Domestic slope results (page rpwcatch ref. G37-G44) and the GOA deep gullies results, for 1990 to 1995 (page rpwcatch ref. G64-G69)

2.2.1.2

These survey indices on the spreadsheet are similar to but not always identical to those given in table 4.2 of Fujioka (1995). However, the differences are probably insignificant. .

The combined results are stored. (page rpwcatch ref. F64-F69)

2.2.1.3

These results are calibrated to absolute biomass using conversion factor ascribed to Rose (1986) (NOT AVAILABLE TO ME) which uses the Clausen and Sigler (1989) (C&S) computation of the cooperative survey series (see table 4.2 of Fujioka (1995)). The factor is.

$$141.4 * \frac{\{0.5[C\&S(1994)+GOA\ Slope(1994)]+GOA\ deep\ gullies(1994)\}}{\{GOA\ Slope(1994)+GOA\ deep\ gullies(1994)\}}$$

This is presumably an attempt to calibrate the domestic series to the absolute biomass in a similar way to that used else where. The logic of this conversion is not immediately apparent to us. It presumably stems from the fact that the domestic GOA slope survey is already calibrated to the cooperative survey (CHECK). Even if it is correct it would seem at least superficially better to estimate the conversion over more than one years results.

The conversion is available at (page rpwcatch ref. F2).

The converted domestic survey series results are divided by 1000 (to bring to 1000 mt.?) and augmented by the pre-survey catch (page rpwcatch ref. H87-H92) divided by 1000.

Results are stored. (page rpwcatch ref. E19-E24)

2.2.2 Gulf of Alaska RWP indices 1979 to 1989. (page rpwcatch ref. D8-D18)

2.2.2.1

Results from 1983-1989 were derived as the analysis of Kimura and Zenger (199?) and appear on the sheet at (page rpwcatch ref. H28-H38)

(INPUTS NOT CHECKED)

Results for 1979-1982 were derived from the Clausen and Sigler (1989) (C&S) computation of the cooperative survey series and calibrated to equivalence with the Kimura and Zenger series using the average ratio of these series from 1983-1994. The complete 1979-1989 series was augmented, by the average proportion the deep gullies results had formed of the GOA slope indices from 1990 -1995. This was to increase these results to form an estimate of both the slope and gullies components of the GOA, so as to be equivalent to the post 1990 results.

The results are stored (page rpwcatch ref. E53-E63)

2.2.2.2

The results described in sub section 2.2.2.1. were calibrated to the total biomass for the GOA, by applying the steps described in sub section 2.2.1.3, with the same conversion factor.

2.2.3 The 1979-1995 results for the Eastern Bering Sea (EBS)

These are stored at (page rpwcatch ref. B8-B24)

2.2.3.1

The S&C results for the GOA for 1979-1981 were calibrated to equate to the Sasaki (1986) results from the EBS (CHECK) (expressed as percentages of his 1979 GOA index) using the ratio of the 1982+1983 results from both series. These and the extended? results of Sasaki (1986) for the EBS from 1982-1988 are stored (page rpwcatch ref. B28-B37)

2.2.3.2

The above results were then converted by a factor of 1887.02/100 which we presume converts the Sasaki (relative to 1979 GOA) index to the level of the EBS cooperative series for the period 1989-1994??. These are stored together with the later results and an value for the 1995 result derive by calibrating the GOA result.

(page rpwcatch ref. B53-B69)

2.2.3

The above results are calibrated to the total biomass using a factor of 141.4 (page rpwcatch ref. F3) divided by 1000 and augmented by the pre-survey catch also divided by 1000 to express results in 1000 mt. The pre-survey catches for the years 1979-1981 were adjusted from the total EBS catch using the ratio of pre-survey to total catch that obtained in 1982-1983.

Results are stored. (page rpwcatch ref. B8-B24)

2.2.4 The 1979-1995 results for the Eastern Bering Sea (EBS)

These are stored at (page rpwcatch ref. C8-C24)

2.2.4.1

Step essentially as 2.2.3.1 above but only 1979 and 1980 results needing conversion from the S&C results.

The results stored. (page rpwcatch ref. C28-C37)

2.2.4.2

Step as 2.2.3.2 above and stored (page rpwcatch ref. C53-C69)

2.2.4.3

Step as 2.2.3.3 above and stored (page rpwcatch ref. C8-C24)

2.2.5 Summary of audit of calibration of the RPW index.

An audit was carried out of the spreadsheet used to construct input data for the 1995 DD assessment. The following points were noted.

1. The RWP index suffers from not being based on one standardized survey for all years and areas. Results therefore have to be calibrated between the various series.
2. Calibrations are made between various constructed series and use various seemingly ad hoc conversion factors. We were not able to check all conversions(LEE IF IN PARTICULAR YOU COULD GET SOME ONE TO EXPLAIN OF POINT OUT THE REFERENCE TO THE SURVEY INDEX TO TOTAL BIOMASS CONVERSION I THINK THAT WOULD BE IMPORTANT)
3. It is probably time that a recalibration was made going back to original data rather than constructs.
4. Inputs that could be checked with published series were generally similar but not always the same. Results shimmer a little but we do believe this changes overall results seriously.

Despite the above caveats we did not find anything (YET) to suggest that the RPW index was flawed.

3 Management Policy

3.1 Introduction

It was noted in the introduction that much of the change in the preliminary 1997 ABC was a result of the management policy shifting from the 35% rule to the 40% rule. That is to say changing to a rule which allows a smaller exploitation proportion. This is a change from what would in most fisheries be a conservative policy to an even more precautionary one. This change is one that the industry should not necessarily oppose if it brings a promise of higher and more consistent yield over time and added to the value of IFQ certificates, even if it requires short term losses. However, it would be proper for the industry to seek the assurance that this is the reasonable expectation of the change. It is not clear that calculations of the reasonable expectations of fishermen have been made except partially for the first tier of the proposed revision. The industry may therefore wish to ask that management agencies underatake such studies. Such investigations, which require large scale simulations, are clearly beyond the scope of this study.

Since we understand that the change in rules for setting the ABC has been agreed by the Fisheries Council at its June 1996 meeting there is probably little point in arguing for the old rule to stand. (LEE WOULD THERE BE ANY POINT IN THE INDUSTRY ARGUING FOR A PROGRESSIVE IMPLEMENTATION?) However it is worth examining the new ABC and OFL rules in detail and seeing how they might apply.

3.2 Current implementation of the new rules.

The new rules are set out in six tiers set out in descending order of preference, corresponding to descending order of information available. In general we would expect that the ABC set under a lower tier would be less than that set under a higher tier. The preliminary sablefish estimation appears to have been set under the third tier. This is when the information available are reliable point estimates of current biomass(B), $B_{40\%}$ =Biomass at 40% of the estimate of average unexploited biomass, and the Fishing Mortality rates ($F_{40\%}$ and $F_{30\%}$). Fishing Mortality rate is a theoretical concept which relates to the exploitation proportion. $F_{40\%}$ and $F_{30\%}$ are the fishing mortality rates which , if applied over a long time period, would on average reduce the biomass to 40% or 30% of its unfished level.

Given this information the third tier rules are as follows.

3a If the current stock is above $B_{40\%}$, then F_{OFL} (the level of fishing mortality rate at which over-fishing is considered to take place) is set equal to $F_{30\%}$ and F_{ABC} (the level of fishing mortality rate which determines the exploitation proportion for the ABC) is set at less or equal to $F_{40\%}$.

3b If (as is currently the case for sablefish) the current biomass is less than $B_{40\%}$, but greater than an arbitrary lower limit, set at 5% of $B_{40\%}$, then F_{OFL} and F_{ABC} are set according to the following formulae:

F_{OFL} equals $F_{30\%}$ *(B/ $B_{40\%}$ - 0.05)/0.95

F_{ABC} is set at less or equal to $F_{40\%}$ *(B/ $B_{40\%}$ - 0.05)/0.95

The effects of these two formulae are to progressively reduce F_{OFL} and F_{ABC} downward, from $F_{30\%}$ and $F_{40\%}$ respectively, by a proportion that is slightly greater than the proportion that biomass forms of $B_{40\%}$.
It is these two formulae that determined the lower levels of the preliminary OFL and the ABC catch levels for 1997. Given the current state of the sablefish stock these rules (3b rather than 3a) are most likely to determine the final OFL and ABC for 1997. Realistic adjustments to the assessments (see section 4) are not likely to change the rule set used.

3.3 Possible implementation at other tiers of the rules

It is worth asking if the third tier is the most appropriate one on which to base the 1997 OFL and ABC. To keep things as simple as possible, we will only examine the effects of adopting the next lower tier of rules (the fourth tier) or the next higher tier of rules (the second tier)

The fourth tier

The fourth tier sets F_{OFL} equal to $F_{30\%}$ and F_{ABC} to less or equal to $F_{40\%}$. It thus seems superficially less restrictive than the third tier.. However, it seems unlikely that the SSC would allow a larger ABC against an argument that less information exists than the assessment scientists consider to be the case. Indeed the thrust of the new measures, (and also more generally of the precautionary approach to fisheries management) is that the benefit of any doubt should be given to the fish and not the fishers. Thus to argue for the 4th tier being appropriate would probably be to argue for a lower ABC.

The second tier.

This tier requires not only the information need for the third tier but also reliable point estimates of F_{MSY} (the level of fishing mortality rate which , if applied over a long time period, would on average give the maximum sustainable yield) and an estimate of B_{MSY} (the average biomass that would occur with this fishing mortality rate).

The average yield that would occur at any level of fishing mortality is usually calculated as the product of two components. These are:-

1. the average amount of yield (usually called the yield per recruit) that could be expected to accrue over time, from one fish joining the stock (recruiting) at some specified age (age 2 in the case of sable fish), and subject to the particular fishing mortality rate.
2. the average numbers of fish that might be expected to recruit at the particular level of fishing mortality rate. This may vary with fishing mortality rate because average levels of the spawning stock biomass might be expected to change with exploitation level and recruit levels might be expected to change with the level of spawning stock biomass. Such a relationship between spawning stock and recruitment is usually called the stock-recruitment relationship.

The product of these two factors predicts the average yield likely to occure as a consequence of a particular level of fishing mortality being applied to the stock. This can be examined at various level of fishing mortality to discover which gives the *maximum sustainable yield* .

The calculation of yield per recruit is straight forward given the results of the age structured assessment model used for sablefish in 1996. The procedure is analogous to that used to calculate $F_{30\%}$ etc. by NMFS scientists. Figure 3.1 (the upper convex curve) shows the calculated yield per average numbers of recruits curve for sablefish, given the results of the current NMFS age based assessment. Yield per average recruit is shown rather than yield per recruit, because this gives a value which is more easily interpreted as analogous to current yield. Maximum yield per recruit on this curve occurs at a level fishing mortality of about 0.65 or at an exploitation proportion considerably larger than the current level at about 48%. If recruitment remained constant at all levels of fishing mortality rate, this curve would also estimate the maximum sustainable yield. However, there is a risk that at such comparatively high exploitation levels, the recruitment might fall because the spawning stock would be eroded.

The relationship between recruitment and stock size in sablefish has not yet become clear. It is possible to plot the recruitment since 1981 against the spawning stock that generated it. This is shown in figure 3.2. No very clear relationship can be seen in the data and thus it is difficult to estimate a true stock recruitment relationship. If anything, the recruitment appears to be larger at the lower spawning stock level observed but this is probably an artifact. NMFS scientists have assumed in their calculations of $F_{30\%}$ etc., that at least above the level of $B_{30\%}$, the recruitment is constant at a level of 17.8 million 2 year-old fish. If spawning stock biomass declined below that level we would eventually expect to see a down turn in recruitment.

A conservative assumption would be to assume that recruitment would start to decline linearly when biomass became lower than the lowest spawning biomass level observed so far(134,000mt). With this rather conservative assumption, we obtain the stock recruitment relationship drawn on figure 3.2. If this is adopted then we can revise the yield per average numbers of recruit curve in figure 3.1. This revised curve is also shown in figure 3.1 (the rising limb of the previous curve plus the concave tail). This curve takes the assumption of declining recruitment into account and so provides a rather conservative total yield This has a maximum near to $B_{30\%}$, $F_{30\%}$ which thus might be taken as conservative estimates of B_{MSY}, and F_{MSY}. Strictly the 2nd tier require point estimates of the values but in providing conservative estimates we are obeying the spirit, if not the letter, of the new regulation. If this was accepted then since the current spawning stock biomass (149,000mt) is higher than $B_{30\%}$ the rule 2a would apply. This is that:-

F_{OFL} is equal to F_{MSY} *($F_{30\%}$ / $F_{30\%}$)=.25 which equates to a exploitation proportion of 22%.

F_{ABC} is set as less or equal to F_{MSY} which equals 0.163 which equates to an exploitation proportion of about 15%.

The later exploitation level is considerably higher than that currently proposed for the 1997 ABC. In practice, how much lower, than 15% level, the ABC exploiation level might be set, would be a matter for the SSC. In practice this line of argument at least holds some prospect of increasing the exploitation proportion somewhat for the 1997 ABC while staying within the revised management guidelines. At worst it should be no worse than the current proposed approach.

Figure 3.1 Yield per average numbers of recuits

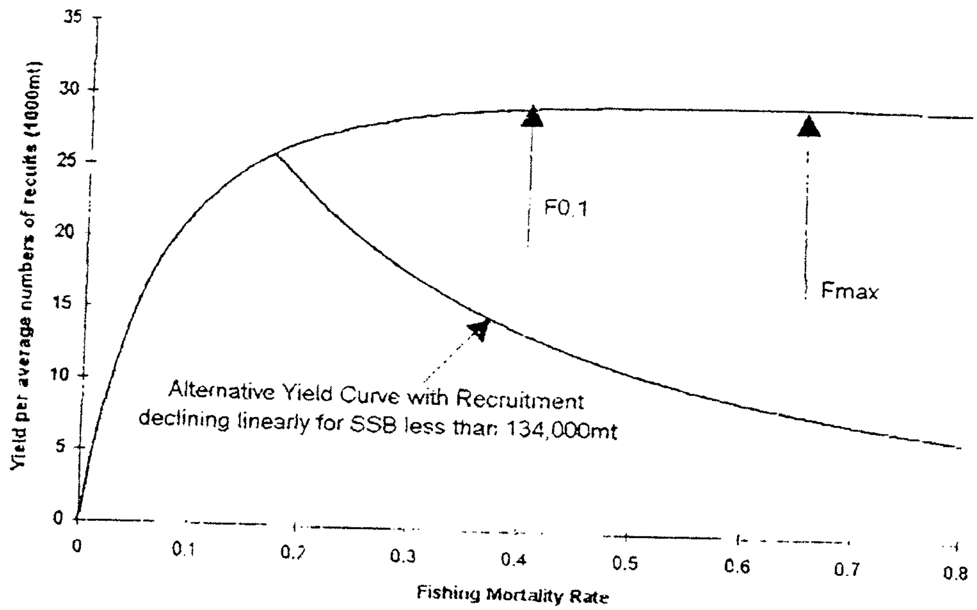
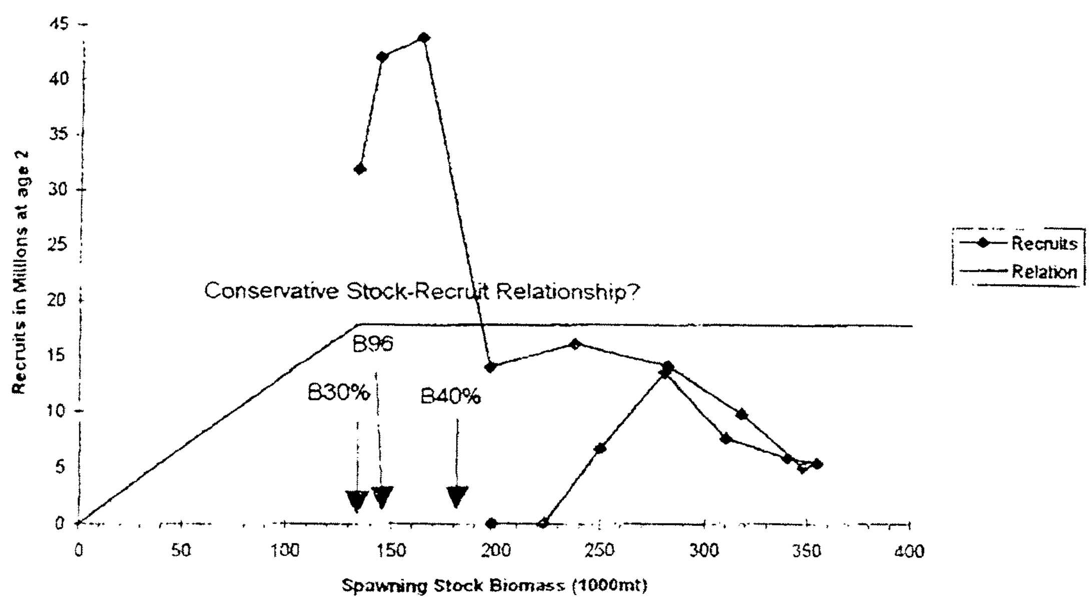


Figure 3.2 Stock Recruitment Plot for sablefish



4. The 1996 Assessment method

The age based model.

The age based assessment method used to estimate the preliminary 1997 sablefish ABC is described in Fujioka *et al* (1996) and implemented in the EXCEL spreadsheet AGESTR~1.XLS.
The model resides in the worksheet called model . It is an integrated age based analysis. That is to say it relies on a minimalist model of the age structured stock dynamics which is fitted to the available survey data by statistical techniques.

A brief description of the model is as follows:-
The age structure model assumes that fishing mortality can be separated into an age specific partial exploitation pattern (catchability by age) based upon a flexible three parameter curve and annual level of fishing mortality. Any given set of these parameters together with parameters of year-class strength and estimates of weight at age, commercial catch and natural mortality allows the sablefish stocks population at age to be estimated at the beginning of the each year and also at the time that the long-line surveys are conducted each year.

To estimate the parameters, commercial catch number (converted from commercial catch weight using average weight data) are used together with an estimate of exploitable population size to estimate fishing mortality (or rather the related estimate of the exploitation proportion) for each year. Total commercial catch numbers data are taken to be exact. The results of the simulated stock structure are then converted to estimates equivalent to the scaled RPN of the long line surveys, the available proportion of the stock at each age in each year (for years where this is available), and the available proportion of the stock at each length of males and of females treated separately. These latter estimates are prepared using matrices of length at age together with the available stock structure at age. These simulated estimates are compared to the real observed outcomes of the long-line surveys and the parameter estimates are adjusted (using the solver tool in EXCEL) to produce the best fit possible. Goodness of fit between the estimates and the observed data are made on the basis of maximum likelihood.

Having obtained best fits to the parameters the model can then be used to estimate:-

1. The unfished level of spawning stock based upon average year-class strength and a given maturity at age schedule and the levels of exploitation that would give 30%, 35% and 40% of the unfished spawning biomass.
2. The catch in the following few years to provide ABC's and OFL levels according to the various management rules.

Consideration of the models structure and assumptions made in its formulation.

The model used seems logically coherent and is in fact one of the few age based approaches possible given the sparse nature of available data from this stock. The spreadsheet implementation, while clearly provisional, appeared to has have been constructed in a professional fashion and was as far as we could ascertain free from programming errors.

We considered whether more detailed and better known age based models (such as *ad hoc* tuned VPA's) might be adapted to this stock but found that the data would not adequately support this approach.

The general class of catch at age based techniques tend to have the virtue that they predict the past absolute stock size (on the basis that when you have caught the lot you know how many there were!). They tend to converge towards these accurate past estimates at a rate dependent upon the level of mortality (i.e. how long it takes to catch the lot). Given the low mortality of sablefish this would argue for relatively slow convergence in the current model. Moreover, given that the current model only uses total commercial catch numbers data this property is not certain. However, we consider that the consequent fits to the age (or size) proportionate structure of the stock should secure this effect. Check Kimura ref.

A virtue of the age based approach is that it avoids the need for having the estimate of total biomass in the sea (estimated by comparisons of long-line surveys and trawl swept area estimates of abundance) that is needed by the delay difference model previously used. A problem is that current estimates of stock size depend on available mortality or stock trend data. Hence the estimates of current stock size will largely depend mostly on the last few values of the survey or other trend data available (in this case the results of the long-line survey).

In the details of the model we note approximation that an approximate form of the Baranov catch equation has been used but that this is reasonable given 1) the low levels of mortality considered, 2) the distribution of fishing in the year. We also note that for those years for which available stock proportions at age are available it may be redundant to also fit the size data.

Given the sparse nature of the available sablefish data ~~the model necessarily has~~ to make a number of assumptions. These are are needed to allow for the model to be used. assumptions are that

1. The annual commercial catch numbers data are known exactly.
2. The commercial catch and the long-line survey both have the same age and size structure in a year.
3. The males and females form the same proportion of the population of each age in each year.
4. Natural mortality is constant with age and time.
5. That catchability at age forms a smooth function of a certain functional form
6. That the degree to which faith can be put in the various data sets is known.

Given the lack of data it is not possible to properly test these assumptions.
We note however that with respect to them:-

1. It is possible that catch was possibly under-registered in years prior to the development of a discard observer program and that losses due to killer whales. long-line drop-offs etc. might be a constant source of additional unrecorded catch related mortality. This is thus one area where the details of the current model might be probed.
2. Commercial fishermen might concentrate on particular ages or sizes of fish while the survey attempts to sample all ages equally. Thus, differences might occur between the survey and the commercial size and age distributions. While it is not possible to test this assumption at present, it should certainly be addressed in the future with suitable sampling of commercial catches.
3. This problem is inherent in the current form of the data. If sex based differences in mortality or availability seem likely to be important it would be better to work with the proportion of each sex at each age or size. ~~which~~ such data are presumable available in the archives of the survey results. In the mean time it is difficult to address this problem.
4. The assumption of constant natural mortality with age and year is a common feature of many fisheries assessment models. It has to be taken as a reasonable working assumption until more detailed predation based models (and their attendant data) can be developed.
5. This seems a reasonable simplifying assumption.
6. The degree of faith or weight to put on each of the three fitted data sets is difficult to establish. Pragmatically, like Fujioka *et al* (1996) we prefer to investigate the effects of different weights. We further consider that, given the differences between the age based model and the delay difference model, it would be worth fitting the age based model to the RPW series rather than to the RPN series since this might help generate some coherence between the approaches..

Hence, given the nature of the data we accept the need to make assumptions 3-5. We suspect that assumption 2 may be important but as yet no data for the scale of this problem are available to us. ~~and~~ We are thus not able to investigate it meaningfully. We therefore ~~thus~~ concentrate our investigations on the effect of the assumption 1) to do with under registered commercial catch and 6)the relative weighting of the three fitted data sets and the choice of survey series.

Investigations of the possible effects of assumptions 1 and 6 on the biomass estimates and the catch forecast for 1996 and preliminary and tentative estimates for 1997.

Various model runs were made using modification of the NMFS spreadsheet AGESTR~1.XLS to question assumptions 1 and 6 above.

Run 1

Run 1 was a basic rerun of the NMFS model starting from different initial conditions. The results while not exactly the same were similar for all reasonable purposes and are included as a baseline. Results from this run are shown at figure 4.1.1-3. Figure 4.1.1 shows the 1996 exploitable population together with the exploitation percentages and the corresponding catch estimates for the 30% adjust. (OFL), the 35% adjust. ABC and the preliminary (based upon 1996 population) 40% adjust new rule ABC. Results for the age based method are (in black) and contrast ed with the equivalent delay difference results (in white). Apart from rounding errors these results are essentially the same as those given in figure 1.

Figure 4.1.2 shows the equivalent results for 1997. Since these are not based upon the 1996 commercial catch and survey results. **WE STRESS THAT THESE RESULTS (AND EQUIVALENT RESULTS FOR FURTHER RUNS) ARE ILLUSTRATIVE AND TENTATIVE** and will certainly change as a result of the 1996 data being included in future analyses. We therefore do not recommend them as a basis for making commercial decisions.

Relative population weight

compares trend

Figure 4.1.3 ~~shows~~ the ~~fit~~ of the scaled available biomass estimated by the model to the RPW survey estimate of total biomass. It will be noted that the model estimate of biomass is lower than the RPW figure in the three most recent years. This difference is the main cause of the discrepancy between the age based and the delay difference estimate of the preliminary 1997 ABC. The figure also gives the estimate of the catchability of the survey series (i.e. the ratio between the survey and the model estimate of available biomass at the time of the survey. This is 1.20 and suggests that the survey calibration to total abundance (via swept area estimates) may have resulted in over estimates of the RPW series (since generally we would not expect swept area catchabilities to exceed 1). Alternatively it might suggest that the model was underestimating the available population. Possible reasons for this might perhaps be that catch data were underestimates or that the wrong level of natural mortality was adopted.

parameters were

will

The model ~~was~~ fitted giving an equal weighting factor ~~to~~ each of the data sets. The goodness of fit measures were 0.165 for the RPW index, 0.268 for the proportion at age data and 1.367 for the proportion at size data. The total goodness of fit measure, which the parameters are chosen to minimize, (1.800= 0.165 + 0.268 + 1.367) is dominated by the proportion at size data. It thus seem worth either down weighting this or up weighting the survey index to see if this will result in a closer fit to the survey index, particularly in the more recent years. This forms the basis of the next run.

Run 2

Run 2 is as run 1 except that the proportion at size data is given a weighting of 0.1 in order to set it to a contribution to the total that is more in balance with the other data sets.
Results equivalent to those for run 1 are shown in figures 4.2.1 to 4.2.3. In particular this modification ~~change~~ results in the following changes relative to run 1.

1. Exploitable biomass estimates are higher in both 1996 and 1997.
2. Exploitable percentages at the 30%,35% and 40% adjust levels are all estimated as larger in both 1996 and 1997.
3. Catch estimates are larger and more comparable to the DD model.
4. Biomass trend in recent years are marginally less steep and catchability marginally lower at 1.19.

The goodness of fit measures were 0.136 (was 0.165) for the RPW index, 0.146 (was 0.268) for the proportion at age data and 1.784 (was 1.367) for the proportion at size data. The total goodness of fit 0.460 measure = 0.136 +0.146 +1.784*0.1 which the parameters are chosen to minimize is now fairly evenly balanced between the results from the three data sets and has allowed the model to adjust better to the RPN and age data at the expense of the size data. This may be better in that we would prefer to have catch at age data given more weight than catch at length data where both exist.

Run 3

Run 3 is as run 1 except that the commercial catch weight ~~has~~ been arbitrarily increased by 10% in all years. ~~each year by 10%~~ to see the effect this has on the model. This is an attempt to see what effects under registering of catches (due to unrecorded catch or discards or non catch fishing deaths such as drop outs or killer whale interference) might have. Results equivalent to those for run 1 are shown in figures 4.3.1 to 4.3.3. In particular this modification ~~change~~ results in the following changes relative to run 1.

1. Exploitable biomass estimates are higher in both 1996 and 1997.
2. Exploitable percentages at the 30%,35% and 40% adjust levels are all estimated as similar to those of run 1 in both 1996 and 1997.
3. Catch estimates are larger and more comparable to the DD model.
4. Biomass trend in recent years are similar to run 1 and catchability is lower at 1.09 by about 10% compared to run 1

The goodness of fit measures were 0.165 (was 0.165) for the RPW index, 0.268 (was 0.268) for the proportion at age data and 1.367 (was 1.367) for the proportion at size data. The total goodness of fit 1.800 measure is thus the same as in run 1. These results indicate that increasing the catch by 10% in all years has scaled the model up to higher biomasses and hence catches but otherwise has left trends unchanged.

Run 4

Run 4 is as run 1 except that the commercial catch weight has been arbitrarily increased each year from 1979 to 1990 by 10% to see the effect this has on the model. Commercial catch weight was however left as recorded for years 1991-1995. This is an attempt to see what effects the under registering of catches, due to the unrecorded discards of earlier years, might have. Results equivalent to those for run 1 are shown in figures 4.4.1 to 4.4.3. In particular this change results in the following changes relative to run 1.

1. Exploitable biomass estimates are slightly higher in both 1996 and 1997 than run 1.
2. Exploitable percentages at the 30%,35% and 40% adjust levels are all estimated as similar to those of run 1 in both 1996 and 1997.
3. Catch estimates are slightly larger than run 1.
4. Biomass trend in recent years are similar to run 1 and catchability is lower at 1.16 compared to run 1.

The goodness of fit measures were 0.157 (was 0.165) for the RPW index. 0.271 (was 0.268) for the proportion at age data and 1.359 (was 1.367) for the proportion at size data. The total goodness of fit 1.787 measure is thus marginally smaller than that of run 1. These results indicate that increasing the catch by 10% in earlier years has caused relatively few changes to the model.

In the previous runs it is noticeable that the scaled estimates of available population are lower than the RPW index in recent years. Generally we would expected a model fitted to trend data such as the RPW to approximate to the index very closely in the latest years. This is because the final years population can be increased or decreased without violating the fit to the catch data. These populations thus tend to be determined by the population trend data. In the previous fits the population trend data was given by the RPN index. This suggests that using the RPW index in the fitting may reconcile the model to this trend index. Since this was used in past models this may increase comparability.

Run 5

Run 5 is as run 1 except that the model is fit to RPW data rather than the RPN data. Results equivalent to those for run 1 are shown in figures 4.5.1 to 4.5.3. In particular this change results in the following changes relative to run 1.

1. Exploitable biomass estimates are higher than run 1 in both 1996 and 1997 and are very comparable to the DD results.
2. Exploitable percentages at the 30%,35% and 40% adjust levels are all estimated as marginally larger than run 1 in both 1996 and 1997.
3. Catch estimates are larger than run and more comparable to, though larger than, the DD model.
4. Biomass trend in recent years is less steep than run 1 and accords well with the RPW trend. Catchability is lower at 1.0 which means the survey and model estimates almost coincide.

The goodness of fit measures were 0.151 (was 0.165) for the RPW index. 0.265 (was 0.268) for the proportion at age data and 1.381 (was 1.367) for the proportion at size data. The total goodness of fit 1.797 measure which the parameters are chosen to minimize is just smaller but broadly comparable to that for run 1. Thus using RPW data has not adversely affected the overall fit and has marginally improved the trend data fit. Thus the RPW data seems as valid as the RPN data for model fitting and may be better in the sense that they have been used in past models and may therefore increase comparability.

Run 6

Run 6 is as run 1 except that the RPW index is used instead of the RPN index and the proportion at size data is given a weighting of 0.1 in order to set it to a contribution to the total that is more in balance with the other data.

Results equivalent to those for run 1 are shown in figures 4.6.1 to 4.6.3. In particular this change results in the following changes relative to run 1.

1. Exploitable biomass estimates are much higher in both 1996 and 1997 than run 1 and higher than the DD model..
2. Exploitable percentages at the 30%,35% and 40% adjust levels are all estimated as higher than in run 1 or the DD model in both 1996 and 1997.
3. Catch estimates are much larger than run 1 and higher than the DD model.

4. Biomass trend in recent years are marginally less steep. The biomass trend in at the 1986 peak was lower and the catchability marginally lower than run 5 at 0.96.

The goodness of fit measures were 0.134 (was 0.165) for the RPW index. 0.138 (was 0.268) for the proportion at age data and 1.755 (was 1.367) for the proportion at size data. The total goodness of fit measure 0.448= 0.134 +0.138 +1.755 *0.1 which the parameters are chosen to minimize is now fairly evenly balanced between the results from the three data sets and has allowed the model to adjust better to the RPW and age data at the expense of the size data. As with run 2, this may be better in that we would prefer to have catch at age data given more weight than catch at length data where both exist.

Summary of Runs

Modifications considered to the basic run shown at run 1 consider changing the weighting of the size data Runs 2 and 6, Changing the catch data, runs 3 and 4, changing the trend data from the RPN index to the RPW index. runs 5 and 6. As might have been anticipated changes in each of these factors increased exploitable biomass and hence catch estimates and decreased the estimate of catchability of the RPW index. As might also have been anticipated, these factors reinforce one another when they are used in combination.

Reasoned arguments might be made for adopting all three changes to the age based assessment.

The relative proportions of stock at size data might be down-weighted because it appears to be the most variable, because it will be most affected by the sex ratio at age assumption and because it in some sense duplicates the more valuable relative proportions of stock at age data. However the proper degree of down-weighting would itself be open to argument.

The commercial catch weight might be plausibly increased to account for under registered landings. However a fixed scaling of catch mostly serves to scale the stock by an equivalent amount. Thus if discards were, for example, a constant proportion of the catch then including them would increase the stock and future catch estimates but leave estimates of the predicted landed catch unchanged unless the practice of discarding could be eliminated.

The RPW index might be adopted as the trend data rather than RPN data because its use would make the model more consistent with the previously used delay difference model and because it relates more closely to catch estimates than does the RPN.

There seems therefore to be reasonable arguments why at least a partial inclusion of the first factor and the third factor might be adopted in future runs of the age based method. Conversely it is of course true that a diligent search for plausible factors that might decrease the catch predictions could also yield results. A likely candidate would be a decrease in the natural mortality estimate used. It is clear, however that results from the age based model can alter as a result of changes in the choice of data sets and the relative weighting they are given in the fit.

It is also worth noting the similarities between the various runs as well as the differences. All agree on the broad size of the exploitable biomass (from 161,000mt to 225,000mt. All agree that current exploitation levels are at about the 10% level and all confirm the general trends in biomass which increased to a peak in 1986-1987 has been decreasing since. The catch predictions for 1997 seem on present data to be more pessimistic than those for 1996 but even more difference in catch prediction is generated by the shift from the old 35% ABC rule to the new 40% ABC rule than by biomass change. There are thus some broad truths that can be utilized to construct sound management policies for the sablefish stock.

4.1.1 etc.

Run , As basic

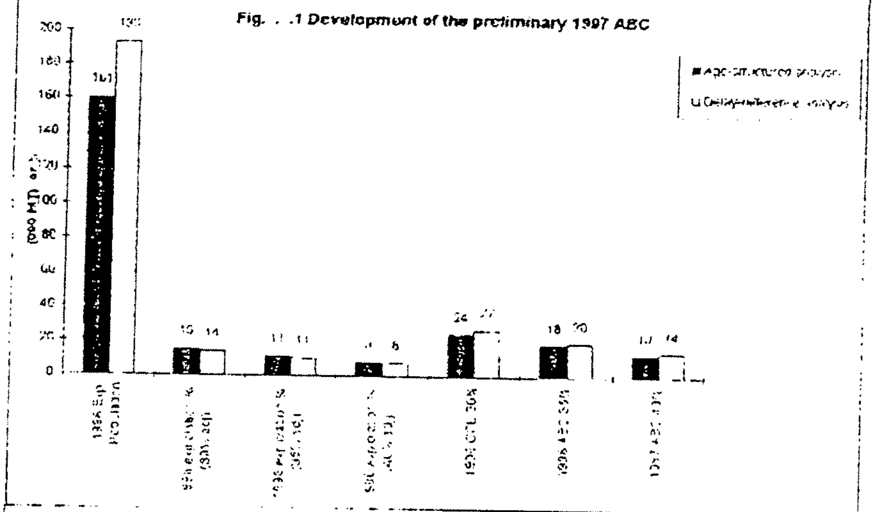


Fig. . .1 Development of the preliminary 1997 ABC

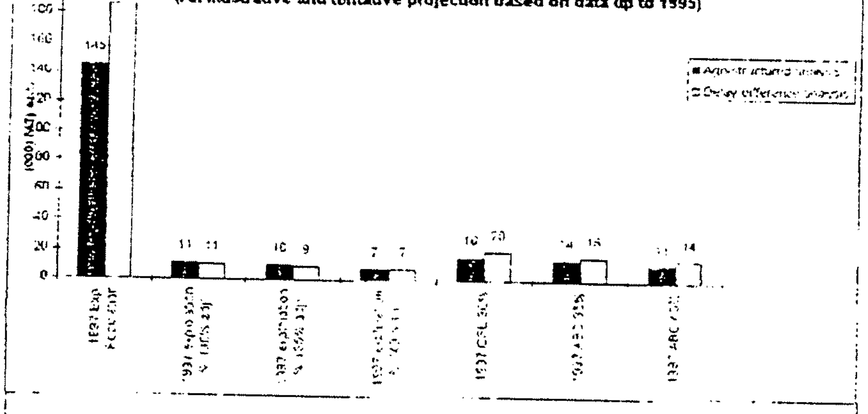


Fig . .2 Development of the 1997 ABC
(An illustrative and tentative projection based on data up to 1995)

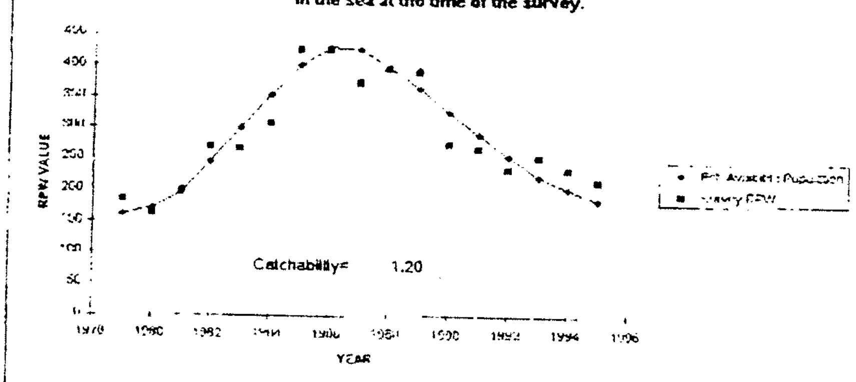


Fig. . .3 Comparison of Survey RPW and scaled estimate of available population in the sea at the time of the survey.

Run . As basic but weights size data by .1

Fig. . .1 Development of the preliminary 1997 ABC

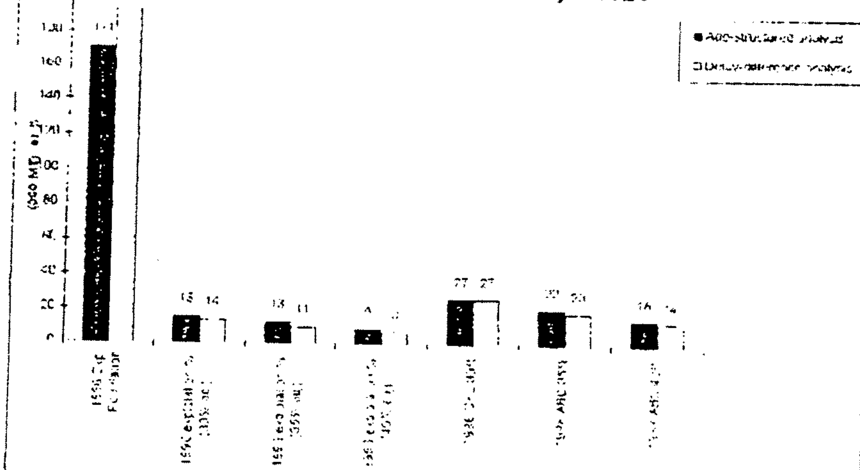


Fig . .2 Development of the 1997 ABC
(An illustrative and tentative projection based on data up to 1995)

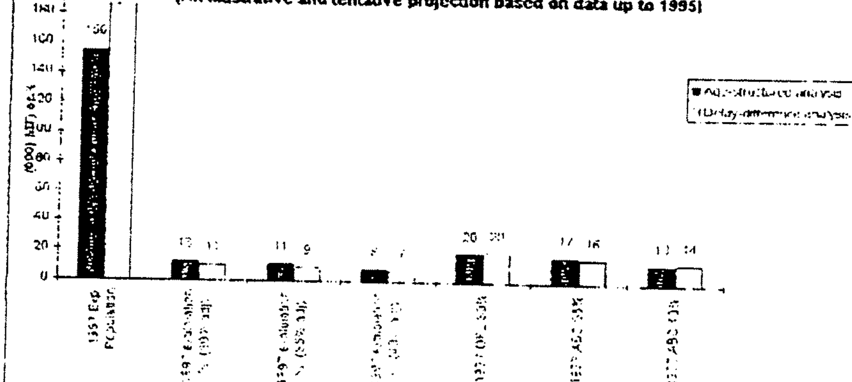
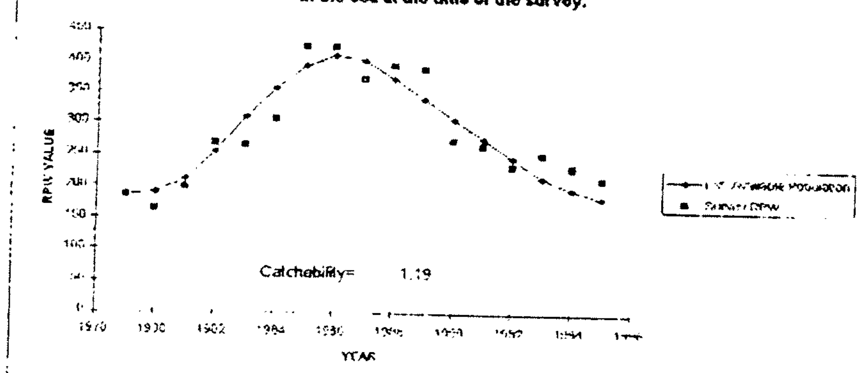


Fig. . .3 Comparison of Survey RPW and scaled estimate of available population in the sea at the time of the survey.



SUMAR16.XLS

Run , As basic but with RPN weighting 1 and catch *1.1 in all years.

Fig. . .1 Development of the preliminary 1997 ABC

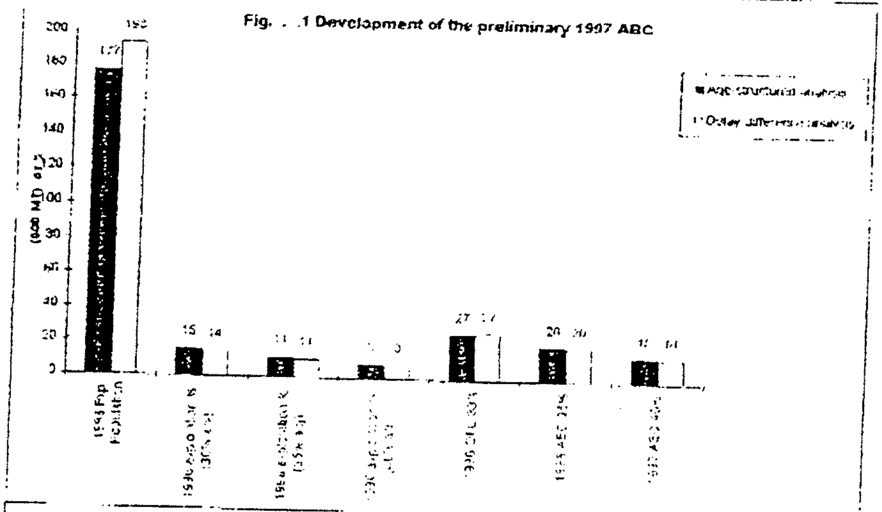


Fig . .2 Development of the 1997 ABC
(An illustrative and tentative projection based on data up to 1995)

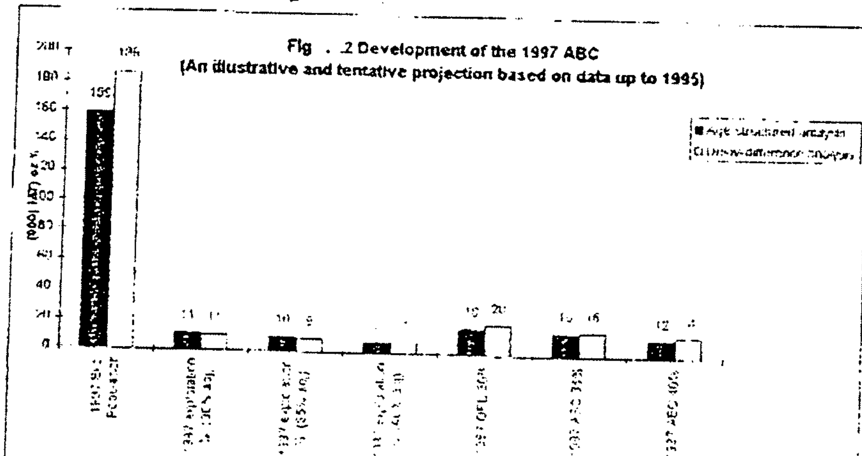
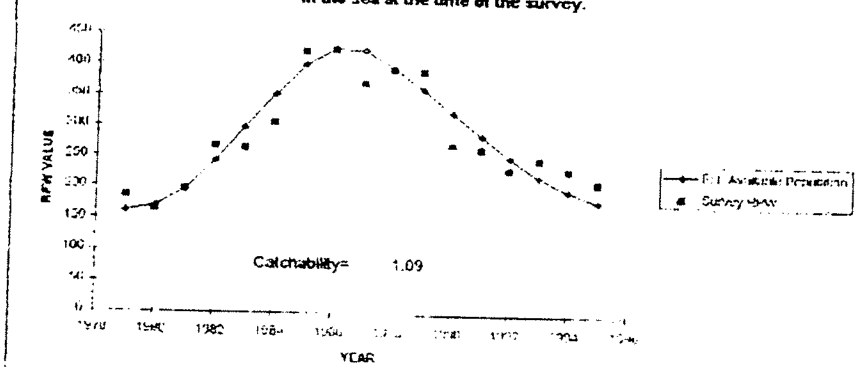


Fig. . .3 Comparison of Survey RPW and scaled estimate of available population in the sea at the time of the survey.



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Run , as basic but with RPN weighting 1 and catch *1.1 in years 1979-90.

Fig. . .1 Development of the preliminary 1997 ABC

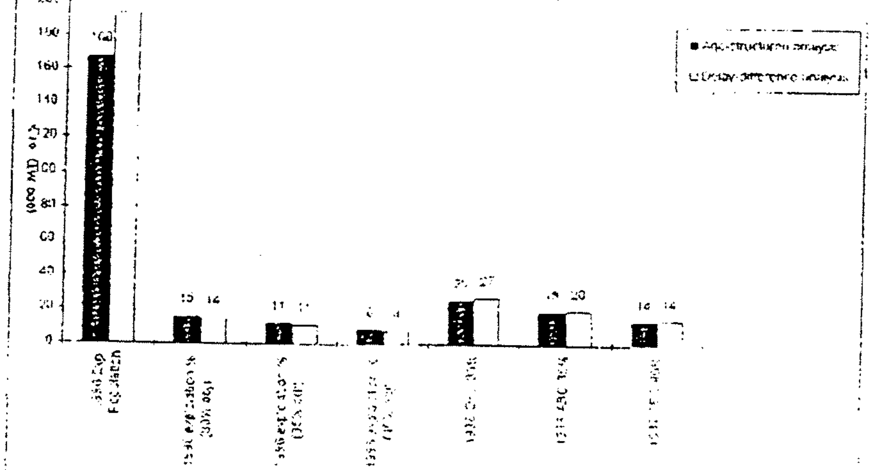


Fig . .2 Development of the 1997 ABC
(An illustrative and tentative projection based on data up to 1995)

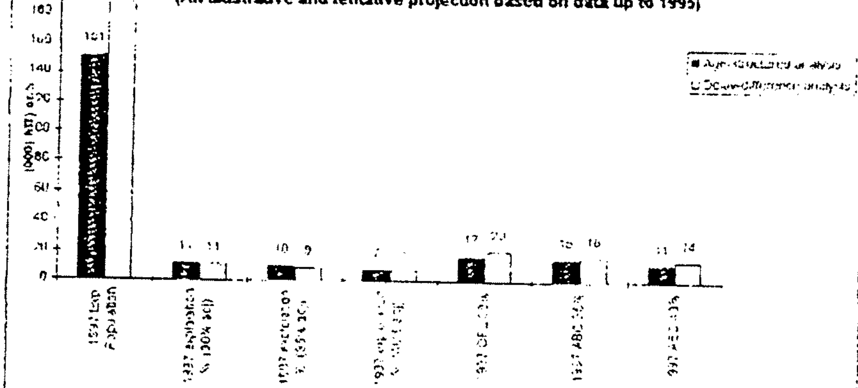
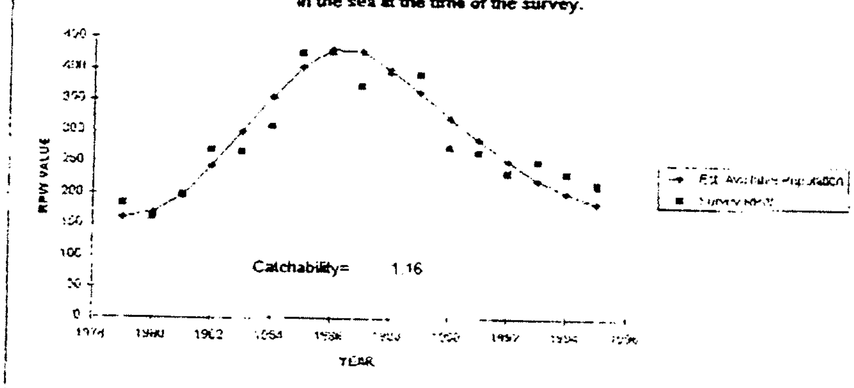


Fig. . .3 Comparison of Survey RPW and scaled estimate of available population in the sea at the time of the survey.



Run , As Basic but fits to RPW rather than RPN.

Fig. . .1 Development of the preliminary 1997 ABC

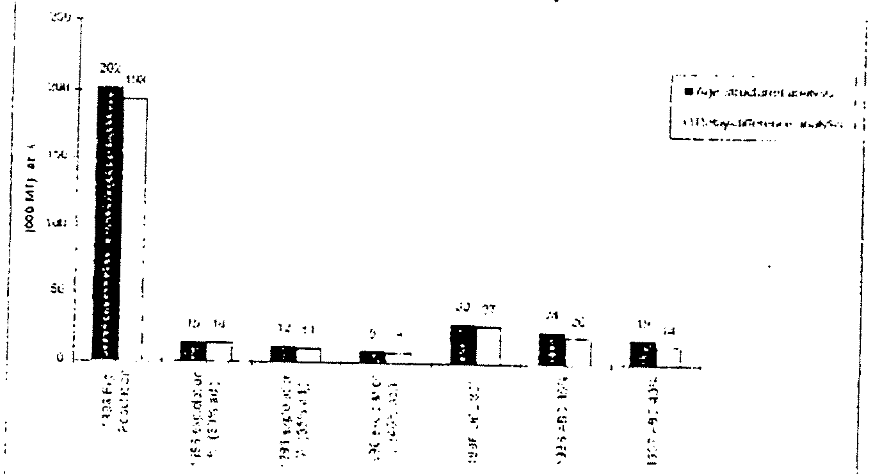


Fig . .2 Development of the 1997 ABC
(An illustrative and tentative projection based on data up to 1995)

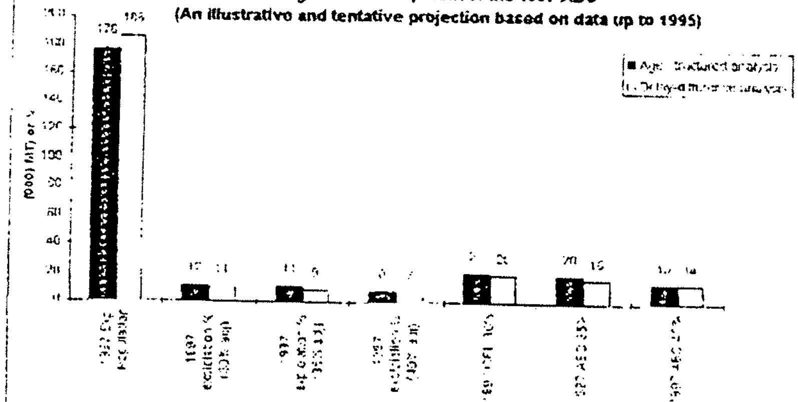
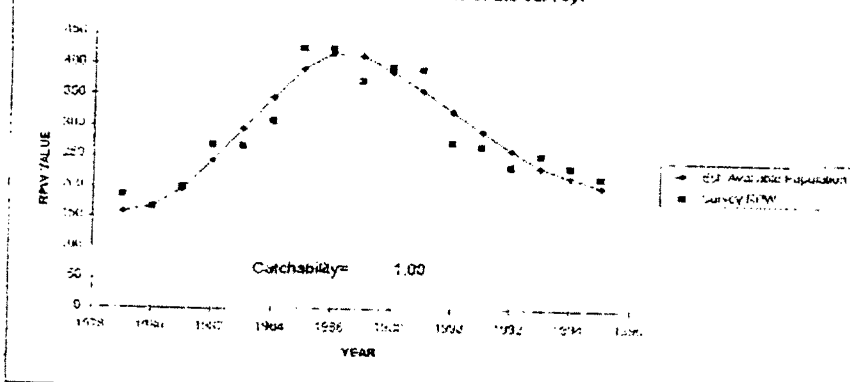


Fig. . .3 Comparison of Survey RPW and scaled estimate of available population in the sea at the time of the survey.



Run , As Basic but fits to RPW and weights size data by .1.

Fig. .1 Development of the preliminary 1997 ABC

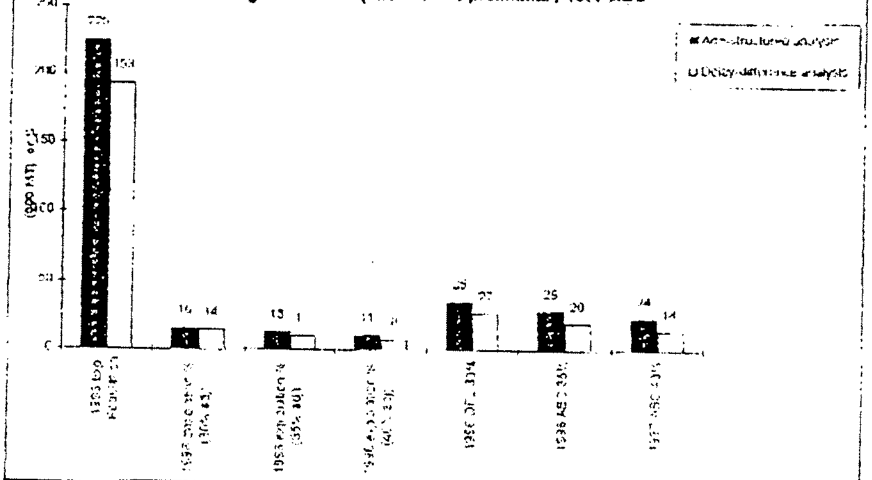


Fig. .2 Development of the 1997 ABC
(An illustrative and tentative projection based on data up to 1995)

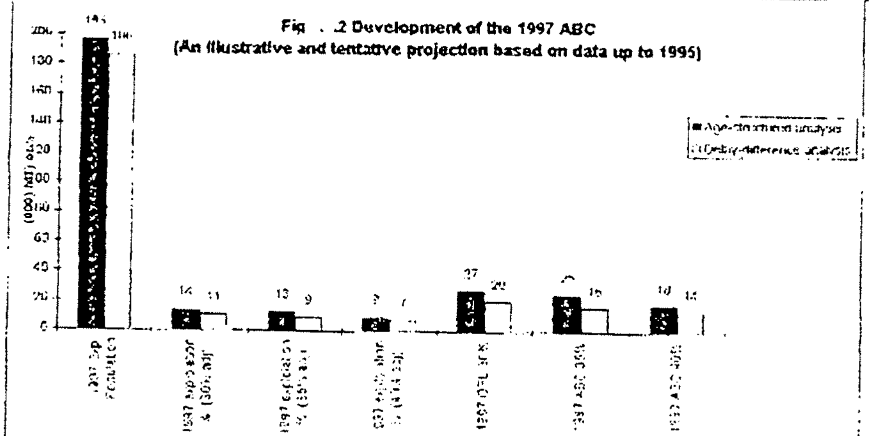
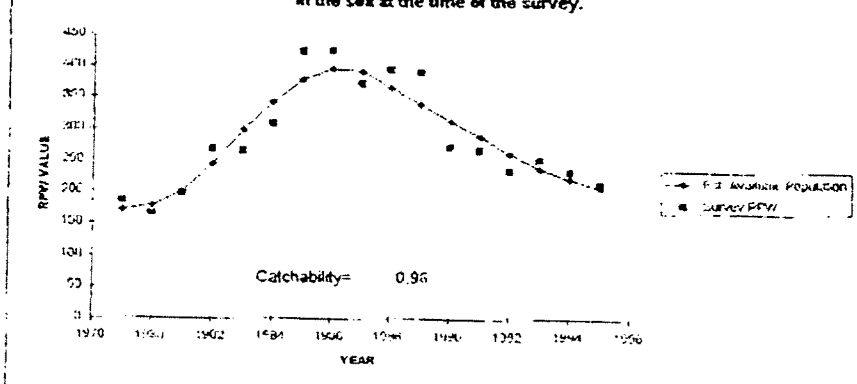


Fig. .3 Comparison of Survey RPW and scaled estimate of available population in the sea at the time of the survey.



Run , As Basic but with RPW and catch *1.1 in all years.

Fig. . .1 Development of the preliminary 1997 ABC

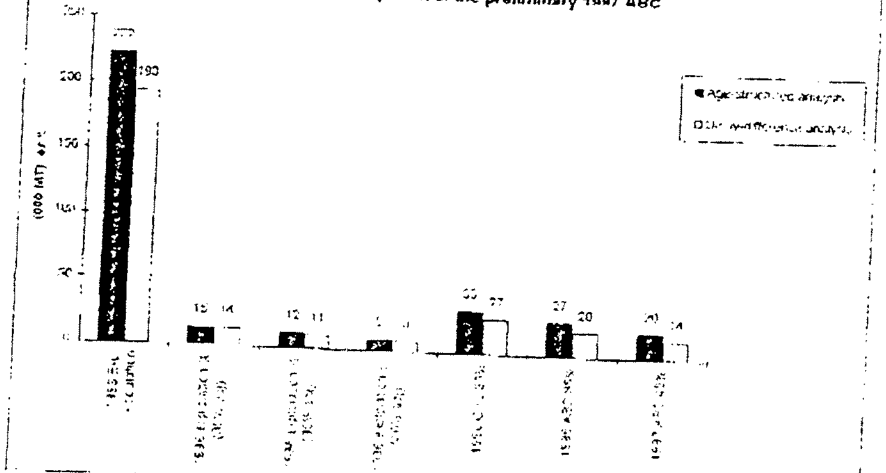


Fig . .2 Development of the 1997 ABC
(An illustrative and tentative projection based on data up to 1995)

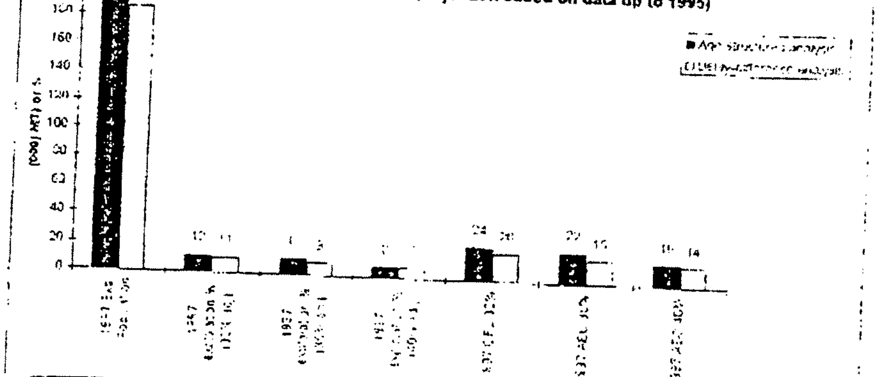
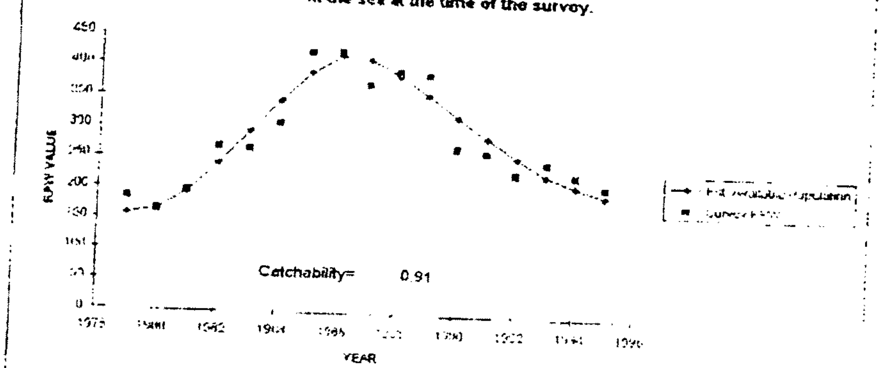


Fig. . .3 Comparison of Survey RPW and scaled estimate of available population in the sea at the time of the survey.



5 Summary and Conclusions.

The preliminary estimate of the 1997 ABC is smaller than the 1996 ABC. This is partly because the biomass is predicted at a lower level by the new age based assessment method, but mainly because new rules for setting ABC's have been adopted which are more precautionary than those used to set the 1995 ABC. It should be noted that the preliminary 1997 ABC was set on the basis of the population in 1996 and thus may change when the assessment is updated in the light of the 1996 survey results and the 1996 commercial catch. Extrapolation from existing trend would suggest that the final estimate may be lower than the provisional estimate.

Examination of the calculations made to calculate the survey biomass trend and the new age based assessment method did not reveal any errors. Rather we conclude they were made in a professional albeit rather preliminary fashion. We believe it might be worth while recalibrating the survey indices from first principles but do not expect that this would show major differences in trend.

Consideration of the new management rules for setting F_{OFL} and F_{ABC} (and hence the basis of calculating OFL and ABC catches), suggests that it would be possible to argue that, with the new age based assessment method, the sablefish might qualify for consideration under the second tier of the rules. This could lead to a more generous interpretations of the OFL and ABC than were made in the preliminary assessment (calculated under the third tier rules).

Reworking the age based assessment with different input assumptions made it possible to reconcile the model predictions more closely with the RPW longline survey index of total stock abundance. In particular, giving less weight to the length distribution data and tuning the results to the RPW, rather than the RPN, increased the compatibility of the new method to the previous approach and resulted in modest but significant increases in ABC and OFL catch levels for 1997.

While a range of interpretations of the age based method exist, all broadly agree on the following points. These are:-

1. On average the RPW index is quite close to the estimated total biomass.

2. Biomass has declined in recent years from the high levels of the mid 1980's

3. The current level of exploitation is comparatively low at about 10%

4. Current levels of spawning biomass are close to the calculated $B_{30\%}$ level.

5. The stock structure is an ageing one due to the low recruitment of young fish in recent years.

The situation of the sablefish stock would therefore seem to be in control but the stock decline is of concern. At present it requires a careful rather than an excessively conservative management approach. Should the biomass continue to decline over the next few years with continuing low recruitments then more stringent measures will be required. The new management rules applied with discretion would seem adequate to this task. The fishing industry should not necessarily oppose reductions in fishing intensity, even though it requires short term losses, if it brings a promise of higher and more consistent yield over time and adds to the value of their IFQ certificates. However, they can reasonably request investigations to establish that the new management will achieve this result. They can also reasonably press for the assessment method and data to be improved to a standard where more optimal exploitation of the sablefish stock can be permitted under the new rules.

Subj: Sablefish
Date: 96-11-08 12:15:50 EST
From: J.G.POPE@dfr.maff.gov.uk (John Pope FSMG DFR)
To: NRCseattle@AOL.COM (Lee Alverson)

Dear Lee I will try to send you corrected text on this paper in a series of emails I am afraid I dont have time to sort out and send the figures but you have the FAX versions.
I will do them in plain text as our gateway tends to screw up anything more complex.
Regards John
There should be 5 messages for the 5 sections to keep things short.

Report on Changes in the Sablefish Assessment.

1. Introduction

1.1. Why do Acceptable Biological Catches (ABC?s) Change?
All estimates of allowable catches rely on two components. These are:-

1. The total weight of [redacted] fish in the sea available for capture by fishermen (We call this the exploitable biomass)
2. The proportion that management proposes, that the allowable catch should form of the exploitable biomass (We will call this the proposed exploitation proportion).

Allowable catches will thus change if either there are more or less exploitable fish in the sea, and if the management agency decides to allow a greater or less proportion of these to be exploited. They may also change, if the methods or data used by fisheries scientists, to estimate the exploitable biomass and the proposed exploitation proportion change.

The second of the two effects can be seen in the changes in the Preliminary Allowable Biological Catches (ABC?s) estimated by NMFS scientists in the Preliminary 1997 Sablefish assessment report of Fujioka et al (1996). This is because the management target is set to change in 1997 from the 35% of unexploited population rule to the 40% of unexploited population rules, for setting proposed exploitation proportions (These rules are explained in the box). The preliminary ABC for 1997 has also changed a result of changing the scientific assessment methodology from the delay difference (DD) method, used in past years, to an age based (AB) method. However, the first effect noted above, change of exploitable biomass, does not yet act on the 1997 ABC. This is because the preliminary 1997 ABC is based upon the 1996 exploitable biomass estimates and not the 1997 exploitable biomass estimates. This is presumably because the results of the 1996 long-line survey results and the commercial catch data for 1996 are not yet available. Therefore, the 1997 exploitable biomass, and hence the 1997 ABC, may change when these new data become available. We may thus ask why the preliminary 1997 ABC has changed and we could also ask how this might change further when the final estimates are available? We might also ask what were the relative importance of the three factors; exploitable biomass change; management target changes for exploitation proportion; changes in scientific methodology; to the changes in the 1997 ABC?

1.2. What was the relative importance of the factors that caused changes in the Preliminary 1997 ABC estimate.
Figure 1 shows the 1996 exploitable biomass, the exploitation proportions that would satisfy the 35% of unexploited population rule and the 40% of unexploited population rule, and the resulting ABC?s associated with these two rules. These results are shown both for the assessment based upon the delay difference analysis (DD) used in previous years and the age based analysis (AB) used in the current assessment. The figure shows that the 1996 estimate of exploitable biomass varied between the two assessment methods. That for the AB was smaller at 162,000 mt than the DD at 193,000 mt. ~~This~~ difference, by itself, would tend to give allowable catches that were 16% lower if the AB method was used compared to the DD method.

A compensating difference, however, occurs when we look at the proposed exploitation proportions associated with the 35% of unexploited population rule. The exploitation proportion estimated by the AB method (11.4%) is 8% larger than that for the DD method (10.6%). A similar 8% difference occurs in the proposed exploitation proportions associated with the 40% of unexploited population rule; 8.3% for AB and 7.6% for DD. Putting together the changes in the exploitable biomass and the changes in the proposed exploitation proportions, result in ABC?s based upon the AB method which are 5% lower for the 35% rule (18500 mt.) than the equivalent result for DD (19600 mt.), and 5% lower for the 40% rule (AB=13300 mt. and DD=14000 mt.).

In these calculations the 35% rule result for the DD method (19600 mt) is effectively (apart from minor rounding errors) the 1996 ABC. In the preliminary calculations of the 1997 ABC, as calculated by the DD method, this has dropped to 14000 mt (i.e. by 28%). This is due to the change of the management target for the exploitation proportion, from the 35% rule to the 40% rule. It drops a further 4% because the Preliminary Assessment, based upon the 40% rule, was estimated by the AB method rather than by the DD method. The perceived change from the 1996 ABC to the predicted 1997 ABC is thus mostly due to the change in management objective from the 35% rule to the 40% rule (28%) but is also due to the change in method (4%). It is not due to a change in biomass as such because both the 1996 (35% rule) ABC and the preliminary 1997 (40% rule) ABC are both based upon estimates of the 1996 exploitable biomass. However, part of the change results from the fact that the perceived size of the 1996 exploitable biomass estimate changes depending on the method and data used to calculate it.
1.3. What might be the relative importance of the factors that may caused changes in the final 1997 ABC estimate.

Ultimately we presume the 1997 ABC will be based upon a 1997 estimate of exploitable biomass (calculated when the 1996 survey results and provisional 1996 catch estimates are available). This can be expected to change from the 1996 level of exploitable biomass. Given recent trends this might be expected to be downward and this could result in a further reduction of the 1997 ABC. Figure 2 shows estimates of the 1997 ABC that are based on biomass projections contained in spreadsheets supplied by NMFS scientists. Readers should note that these are TENTATIVE since they DO NOT include the 1996 survey results and provisional 1996 catch estimates in the assessment and they are included for illustrative purposes only. However the industry should note that the final estimate could possibly be lower than the provisional 13300 mt. estimate.

In Figure 2 it will be noted that the estimate of the 1997 exploitable biomass estimated by DD has fallen by 3% to 186,000 mt. A fall in biomass when the biomass is lower than the target level (see box on 35% and 40% rules) leads to a greater reduction in ABC because the exploitation proportion is then adjusted downward under both the 35% and the 40% rules. Thus the DD estimate of the 1997 ABC under the 35% rule (19,000 mt.) is 3% less than the ABC in 1996. If the 40% rule is adopted (as is planned for 1997) the ABC estimated by the DD method (13,000 mt.) is 29% (i.e. a further 26% lower) than the 1996 ABC. If the AB method of assessment is used then the ABC calculated under the 40% rule (11,000 mt.) is 43% lower (a further 14%). Thus the 43% change in this TENTATIVE 1997 ABC from the 1996 level might be approximately ascribed as being 3% due to biomass change, 26% due to the change in management policy and 14% due to the change in assessment methodology.

1.4. summary and aims

It is clear from the above calculations that the main changes in the preliminary estimate of the 1997 sablefish ABC are due to the proposed change in management policy from the 35% rule to the 40% rule. However, methodology change has had some effect on the preliminary estimate and might possibly (this can only be judged in the light of the final 1996 assessment) be more important in the final assessment of the 1997 sablefish ABC. Changes in the biomass between 1996 and 1997 may also affect the final 1997 sablefish ABC. Current indications are that this may be a fairly minor effect (at least as projected by the DD method) but the perception of this might change in the light of the results of the 1996 long-line survey.

The purpose of this investigation is thus to examine:-

1. Are biomass changes being tracked accurately?
2. What are the changes in the management policy and can the interpretation be modified.
3. Is the new methodology proposed for estimating the 1997 ABC a). Sound? b). has it been used accurately?

Ref

Fujioka et al (1996).

---------------------- Headers ------------------------------

From J.G.POPE@dfr.maff.gov.uk Fri Nov 8 12:15:33 1996
Return-Path: J.G.POPE@dfr.maff.gov.uk

Received: from gatekeeper.maff.gov.uk (gatekeeper.maff.gov.uk [148.252.1.1])
by eminl1.mail.aol.com (8.6.12/8.6.12) with ESMTP id MAA01607 for
<NRCseattle@AOL.COM>; Fri, 8 Nov 1996 12:15:29 -0500
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Sat, 9 Nov 1996 01:13:00 +0000
Date: Sat, 9 Nov 1996 01:13:00 +0000
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From: John Pope FSMG DFR <J.G.POPE@dfr.maff.gov.uk>
Message-ID:
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To: Lee Alverson <NRCseattle@AOL.COM>
Subject: Sablefish

Subj.: Sablefish 2
Date: 96-11-08 12:16:37 EST
From: J.G.POPE@dfr.maff.gov.uk (John Pope FSMG DFR)
To: NRCseattle@AOL.COM (Lee Alverson)

2 Are biomass changes being tracked accurately?
2.1 Introduction
It is noted in the introduction that the level of exploitable biomass is one of two main elements in the calculation of an ABC. It is also noted that actual exploitable biomass change is not directly implicated in the reduced level of the preliminary 1997 ABC relative to the 1996 ABC. This is because the preliminary 1997 ABC is set against the 1996 exploitabe biomass. Some changes in the preliminary 1997 ABC are, however, due to changes in the percieved exploitable biomass which result from a change in assessment method. However, change in actual exploitable biomass between 1996 and 1997 may well influence the level of the final 1997 ABC.

As well as being a measure of what weight of fish is available to catch, the level of the exploitable biomass also determines how the 35% and 40% rules will operate to reduce the proposed exploitation proportion when the Delay Difference (DD). model is used to provide estimates of the ABC. The closely related spawning biomass (the weight of fish available to spawn) has a similar role when the Age Based Assessment method (AB) is used to calculate the ABC. Thus both the level and the trend of the exploitable biomass has a large potentially effect on the estimated level of the ABC and requires tracking accurately. Assessments of the sablefish, by both the DD and the AB models, use abundance indices from long-line surveys as vital data sources for estimating exploitable biomass.

The Delay Difference model (DD) uses estimates of the total sablefish biomass (Relative Population Weight (RPW)) obtained from long-line surveys from the Eastern Bering sea, the Aleutian Islands Region and the Gulf of Alaska. These are calibrated to absolute biomass using comparisons made between long-line survey results and groundfish trawl survey results. The model uses these results as inputs and tracks them exactly. The DD method is thus dependent upon the longline survey index of Relative Population Weight tracking the trend in biomass and also being correctly calibrated to the absolute biomass, if it is to give reliable estimates of fishing intensity and ABC?s.

The Age Based Assessment (AB) method uses estimates of the relative population numbers (RPN) obtained from long-line surveys from the Eastern Bering sea, the Aleutian Islands Region and the Gulf of Alaska.(CHECK ITS FROM ALL AREAS). This is one of the data sets to which the model is calibrated. The model uses the trend in these results but does not need an estimate of absolute numbers. It is thus dependent only on the trend being accurate.

None of the available longline surveys series spans the entire period of the analyses and therefore it is necessary for them to be inter-calibrated. Additionally, in the case of the earlier years in the Bering sea and the Aleutian Island region, missing values have to be made up. In the case of the RPW series, it is also necessary to calibrate indices to absolute biomass.

Since the trends of survey indices influence the estimates of exploitable biomass made by both models it is necessary to check whether the inter-calibration of the separate series has been carried out in a reasonable fashion. Since the absolute level of the RPW series influences the DD model it is also necessary to check how this calibration was made. Finally, as was noted in the introduction, the exploitable biomass for 1996 estimated by the DD method was larger than that estimated by the AB method. The 1996 DD method?s estimate of exploitable biomass is essentially a one year projection from the RWP series. This index is calibrated to trawl survey swept area results (assuming that all sablefish in the path of the trawl were caught), so that it might have been expected to underestimate exploitable biomass compared to the AB estimates (because not all fish in the trawl path might have been caught). The fact that this was not the case seems curious and looks like a loose end that needs examination. The following subsections thus examine the inter-calibration of the survey series in detail and tug at the loose end.

2.2 Inter-calibration of the RPW index.

The inter-calibration of the RWP index is described in Fujioka (1995). Detailed calculations were available to us from spreadsheet DELAYDIF.XLS page rpwcatch. This was kindly made available to us by NMFS scientists. We have conducted an audit of the calculations made on the sheet and these are described below (Lee we probably should move this to a technical appendix). Indices are calibrated separately for the Eastern Bering Sea, The Aleutian Island area and the Gulf of Alaska and then combined. The last area is by far the most important and we deal with its calibration first for two separate time periods.

2.2.1 Gulf of Alaska RWP indices 1990 to 1995. (page rpwcatch ref. E19-E24)

2.2.1.1
These are based upon the Domestic survey series and constructed as follows. The spreadsheet combines the GOA Domestic slope results (page rpwcatch ref. G37-G44)
and the GOA deep gullies results, for 1990 to 1995 (page rpwcatch ref. G64-G69)

2.2.1.2
These survey indices on the spreadsheet are similar to but not always identical to those given in table 4.2 of Fujioka (1995). However, the differences are probably insignificant.

The combined results are stored. (page rpwcatch ref. F64-F69)

2.2.1.3
These results are calibrated to absolute biomass using conversion factor ascribed to Rose (1986) (NOT AVAILABLE TO ME) which uses the Clausen and

Sigler (1989) (C&S) computation of the cooperative survey series (see table 4.2 of Fujioka (1995)). The factor is,

141.4*{0.5[C&S(1994)+GOA Slope(1994)]+GOA deep gullies(1994)}
{GOA Slope(1994)+GOA deep gullies(1994)}

This is presumably an attempt to calibrate the domestic series to the absolute biomass in a similar way to that used else where. The logic of this conversion is not immediately apparent to us. It presumably stems from the fact that the domestic GOA slope survey is already calibrated to the cooperative survey (CHECK). Even if it is correct it would seem at least superficially better to estimate the conversion over more than one years results.
The conversion is available at (page rpwcatch ref. F2).

The converted domestic survey series results are divided by 1000 (to bring to 1000 mt.?) and augmented by the pre-survey catch (page rpwcatch ref. H87-H92) divided by 1000.
Results are stored. (page rpwcatch ref. E19-E24)

2.2.2 Gulf of Alaska RWP indices 1979 to 1989. (page rpwcatch ref. D8-D18)

2.2.2.1
Results from 1983-1989 were derived as the analysis of Kimura and Zenger (199?) and appear on the sheet at (page rpwcatch ref. H28-H38)
(INPUTS NOT CHECKED)
Results for 1979-1982 were derived from the Clausen and Sigler (1989) (C&S) computation of the cooperative survey series and calibrated to equivalence with the Kimura and Zenger series using the average ratio of these series from 1983-1994.
The complete 1979-1989 series was augmented, by the average proportion the deep gullies results had formed of the GOA slope indices from 1990 -1995. This was to increase these results to form an estimate of both the slope and gullies components of the GOA, so as to be equivalent to the post 1990 results.
The results are stored (page rpwcatch ref. E53-E63)

2.2.2.2
The results described in sub section 2.2.2.1, were calibrated to the total biomass for the GOA, by applying the steps described in sub section 2.2.1.3, with the same conversion factor.

2.2.3 The 1979-1995 results for the Eastern Bering Sea (EBS)
These are stored at (page rpwcatch ref. B8-B24)

2.2.3.1
The S&C results for the GOA for 1979-1981 were calibrated to equate to the Sasaki (1986) results from the EBS (CHECK) (expressed as percentages of his

1979 GOA index) using the ratio of the 1982+1983 results from both series. These and the extended? results of Sasaki (1986) for the EBS from 1982-1988 are stored

(page rpwcatch ref. B28-B37)

2.2.3.2
The above results were then converted by a factor of 1887.02/100 which we presume converts the Sasaki (relative to 1979 GOA) index to the level of the EBS cooperative series for the period 1989-1994??. These are stored together with the later results and an value for the 1995 result derive by calibrating the GOA result.

(page rpwcatch ref. B53-B69)

2.2.3

The above results are calibrated to the total biomass using a factor of 141.4

(page rpwcatch ref. F3)

divided by 1000 and augmented by the pre-survey catch also divided by 1000 to express results in 1000 mt. The pre-survey catches for the years 1979-1981 were adjusted from the total EBS catch using the ratio of pre-survey to total catch that obtained in 1982-1983.
Results are stored. (page rpwcatch ref. B8-B24)

2.2.4 The 1979-1995 results for the Eastern Bering Sea (EBS)
These are stored at (page rpwcatch ref. C8-C24)

2.2.4.1
Step essentially as 2.2.3.1 above but only 1979 and 1980 results needing conversion from the S&C results.
The results stored. (page rpwcatch ref. C28-C37)

2.2.4.2
Step as 2.2.3.2 above and stored (page rpwcatch ref. C53-C69)

2.2.4.3
Step as 2.2.3.3 above and stored (page rpwcatch ref. C8-C24)

2.2.5 Summary of audit of calibration of the RPW index.

An audit was carried out of the spreadsheet used to construct input data for the 1995 DD assessment. The following points were noted.

1. The RWP index suffers from not being based on one standardized survey for all years and areas. Results therefore have to be calibrated between the various series.
2. Calibrations are made between various constructed series and use various seemingly ad hoc conversion factors. We were not able to check all conversions(LEE IF IN PARTICULAR YOU COULD GET SOME ONE TO EXPLAIN OR POINT OUT THE REFERENCE TO THE SURVEY INDEX TO TOTAL BIOMASS CONVERSION I THINK THAT WOULD BE IMPORTANT. The ref is to rose 1996 ? I think)
3. It is probably time that a recalibration was made going back to original data rather than constructs.

4. Inputs that could be checked with published series were generally similar but not always the same. Results shimmer a little but we do believe this changes overall results seriously.

~~Despite the above~~ caveats we did not find anything (YET) to suggest that the RPW index was flawed.

2.3 Inter-calibration of the RPN index.

2.4 Comparisons of the exploitable biomass calculated by the delay difference model and the age based model.

---------------------- Headers ------------------------------
From J.G.POPE@dfr.maff.gov.uk Fri Nov 8 12:16:09 1996
Return-Path: J.G.POPE@dfr.maff.gov.uk
Received: from gatekeeper.maff.gov.uk (gatekeeper.maff.gov.uk [148.252.1.1])
by emin28.mail.aol.com (8.6.12/8.6.12) with ESMTP id MAA08671 for
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From: John Pope FSMG DFR <J.G.POPE@dfr.maff.gov.uk>
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To: Lee Alverson <NRCseattle@AOL.COM>
Subject: Sablefish 2

Subj: Sablefish 3
Date: 96-11-08 12:15:51 EST
From: J.G.POPE@dfr.maff.gov.uk (John Pope FSMG DFR)
To: NRCseattle@AOL.COM (Lee Alverson)

3 Management Policy

3.1 Introduction

It was noted in the introduction that much of the change in the preliminary 1997 ABC was a result of the management policy shifting from the 35% rule to the 40% rule. That is to say changing to a rule which allows a smaller exploitation proportion. This is a change from what would in most fisheries be a conservative policy to an even more precautionary one. This change is one that the industry should not necessarily oppose if it brings a promise of higher and more consistent yield over time and adds to the value of IFQ certificates, even if it requires short term losses. However, it would be proper for the industry to seek the assurance that this is the reasonable expectation of the change. It is not clear that calculations of the reasonable expectations of fishermen have been made except partially for the first tier of the proposed revision. The industry may therefore wish to ask that management agencies underatake such studies. Such investigations, which require large scale simulations, are clearly beyond the scope of this study.

Since we understand that the change in rules for setting the ABC has been agreed by the Fisheries Council at its June 1996 meeting there is probably little point in arguing for the old rule to stand. (LEE WOULD THERE BE ANY POINT IN THE INDUSTRY ARGUING FOR A PROGRESSIVE IMPLEMENTATION?) However it is worth examining the new ABC and OFL rules in detail and seeing how they might apply.

3.2 Current implementation of the new rules.

The new rules are set out in six tiers set out in descending order of preference, corresponding to descending order of information available. In general we would expect that the ABC set under a lower tier would be less than that set under a higher tier. The preliminary sablefish estimation appears to have been set under the third tier. This is when the information available are reliable point estimates of current biomass(B), B40% =Biomass at 40% of the estimate of average unexplcited biomass, and the Fishing Mortality rates (F40% and F30%.). Fishing Mortality rate is a theoretical concept which relates to the exploitation proportion. F40% and F30% are the fishing mortality rates which , if applied over a long time period, would on average reduce the biomass to 40% or 30% of its unfished level.

Given this information the third tier rules are as follows.

3a If the current stock is above B40% , then FOFL (the level of fishing mortality rate at which over-fishing is considered to take place) is set equal to F30% and FABC (the level of fishing mortality rate which determines the exploitation proportion for the ABC) is set at less or equal to F40% .

3b If (as is currently the case for sablefish) the current biomass is less than B40% , but greater than an arbitrary lower limit, set at 5% of B40% , then FOFL and FABC are set according to the following formulae;

FOFL equals F30% *(B/ B40% - 0.05)/0.95

FABC is set at less or equal to F40% *(B/ B40% - 0.05)/0.95

The effects of these two formulae are to progressively reduce FOFL and FABC downward, from F30% and F40% respectively, by a proportion that is slightly greater than the proportion that biomass forms of B40%.
It is these two formulae that determined the lower levels of the preliminary OFL and the ABC catch levels for 1997. Given the current state of the sablefish stock these rules (3b rather than 3a) are most likely to determine the final OFL and ABC for 1997. Realistic adjustments to the assessments (see section 4) are not likely to change the rule set used.

3c For completeness, the final set of rules for the third tier set FOFL and FABC and equal to zero if
B is less than 5% of the size of B40% .
However, fortuanatly this rule seems unlikely to be applied in the case of the sablefish at present.
3.3 Possible implementation at other tiers of the rules
It is worth asking if the third tier is the most appropriate one on which to base the 1997 OFL and ABC. To keep things as simple as possible, we will only examine the effects of adopting the next lower tier of rules (the fourth tier) or the next higher tier of rules (the second tier)

The fourth tier
The fourth tier sets FOFL equal to F30% and FABC to less or equal to F40% . It thus seems superficially less restrictive than the third tier..
However, it seems unlikely that the SSC would allow a larger ABC against an argument that less information exists than the assessment scientists consider to be the case. Indeed the thrust of the new measures, (and also more generally of the precautionary approach to fisheries management; is that the benefit of any doubt should be given to the fish and not the fishers. Thus to argue for the 4th tier being appropriate would probably be to argue for a lower ABC.

The second tier.
This tier requires not only the information need for the third tier but also reliable point estimates of FMSY (the level of fishing mortality rate which , if applied over a long time period, would on average give the maximum sustainable yield) and an estimate of BMSY
(the average biomass that would occur with this fishing mortality rate).

The average yield that would occur at any level of fishing mortality is usually calculated as the product of two components. These are:-

1. the average amount of yield (usually called the yield per recruit) that could be expected to accrue over time,from one fish joining the stock (recruiting) at some specified age (age 2 in the case of sable fish), and subject to the particular fishing mortality rate.

2. the average numbers of fish that might be expected to recruit at the particular level of fishing mortality rate. This may vary with fishing mortality rate because average levels of the spawning stock biomass might be expected to change with exploitation level and recruit levels might be expected to change with the level of spawning stock biomass. Such a relationship between spawning stock and recruitment is usually called the stock-recruitment relationship.

The product of these two factors predicts the average yield likely to occure as a consequence of a particular level of fishing mortality being applied to the stock. This can be examined at various level of fishing mortality to discover which gives the maximum sustainable yield .

The calculation of yield per recruit is straight forward given the results of the age structured assessment model used for sablefish in 1996. The procedure is analogous to that used to calculate F30% etc. by NMFS scientists. Figure 3.1 (the upper convex curve) shows the calculated yield per average numbers of recruits curve for sablefish, given the results of the current NMFS age based assessment. Yield per average recruit is shown rather than yield per recruit, because this gives a value which is more easily interpreted as analogous to current yield. Maximum yield per recruit on this curve occurs at a level fishing mortality of about 0.65 or at an exploitation proportion considerably larger than the current level at about 48%. If recruitment remained constant at all levels of fishing mortality rate, this curve would also estimate the maximum sustainable yield. However, there is a risk that at such comparatively high exploitation levels, the recruitment might fall because the spawning stock would be eroded.

The relationship between recruitment and stock size in sablefish has not yet become clear. It is possible to plot the recruitment since 1981 against the spawning stock that generated it. This is shown in figure 3.2. No very clear relationship can be seen in the data and thus it is difficult to estimate a true stock recruitment relationship. If anything, the recruitment appears to be larger at the lower spawning stock level observed but this is probably an artifact. NMFS scientists have assumed in their calculations of F30% etc., that at least above the level of B30% , the recruitment is constant at a level of 17.8 million 2 year-old fish. If spawning stock biomass declined below that level we would eventually expect to see a down turn in recruitment.

A conservative assumption would be to assume that recruitment would start to decline linearly when biomass became lower than the lowest spawning biomass level observed so far(134,000mt). With this rather conservative assumption, we obtain the stock recruitment relationship drawn on figure 3.2. If this is adopted then we can revise the yield per average numbers of recruit curve in figure 3.1. This revised curve is also shown in figure 3.1 (the rising limb of the previous curve plus the concave tail). This curve takes the assumption of declining recruitment into account and so provides a rather conservative total yield This has a maximum near to B30% , F30% which thus might be taken as conservative estimates of BMSY, and FMSY . Strictly the 2nd tier require point estimates of the values but in providing conservative estimates we are obeying the spirit, if not the letter, of the new regulation. If this was accepted then since the current spawning stock biomass (149,000mt) is higher than B30% the rule 2a would apply. This is that:-

FOFL is equal to FMSY *(F30% / F40%)=.25 which equates to a exploitation proportion of 22%.

FABC is set as less or equal to FMSY which equals 0.163 which equates to an exploitation proportion of about 15%.

The later exploitation level is considerably higher than that currently proposed for the 1997 ABC. In practice, how much lower, than 15% level, the ABC exploiation level might be set, would be a matter for the SSC. In practice this line of argument at least holds some prospect of increasing the exploitation proportion somewhat for the 1997 ABC while staying within the revised management guidelines. At worst it should be no worse than the current proposed approach.

---------------------- Headers -------------------------------

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Received: from gatekeeper.maff.gov.uk (gatekeeper.maff.gov.uk [148.252.1.1]) by emin11.mail.aol.com (8.6.12/8.6.12) with ESMTP id MAA01608 for <NRCseattle@AOL.COM>; Fri, 8 Nov 1996 12:15:29 -0500
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Message-ID:
<0007716E.MAI*/I=JG/S=POPE/OU=DFR/O=MAFF/PRMD=MAFF400/ADMD=ATTmail/C=GB/@MHS>
To: Lee Alverson <NRCseattle@AOL.COM>
Subject: Sablefish 3

Subj: Sablefish 4
Date: 96-11-08 12:16:37 EST
From: J.G.POPE@dfr.maff.gov.uk (John Pope FSMG DFR)
To: NRCseattle@AOL.COM (Lee Alverson)

The 1996 Assessment method
The age based model.
The age based assessment method used to estimate the preliminary 1997 sablefish ABC is described in Fujioka et al (1996) and implemented in the EXCEL spreadsheet AGESTR~1.XLS.
The model resides in the worksheet called model . It is an integrated age based analysis. That is to say it relies on a minimalist model of the age structured stock dynamics which is fitted to the available survey data by statistical techniques.

A brief description of the model is as follows:-
The age structure model assumes that fishing mortality can be separated into an age specific partial exploitation pattern (catchability by age) based upon a flexible three parameter curve and annual level of fishing mortality. Any given set of these parameters together with parameters of year-class strength and estimates of weight at age, commercial catch and natural mortality allows the sablefish stocks population at age to be estimated at the beginning of each year and also at the time that the long-line surveys are conducted each year.

To estimate the parameters, commercial catch number (converted from commercial catch weight using average weight data) are used together with an estimate of exploitable population size to estimate fishing mortality (or rather the related estimate of the exploitation proportion) for each year. Total commercial catch numbers data are taken to be exact. The results of the simulated stock structure are then

1. Scaled to estimates equivalent to the Relative Population Number (RPN) index of the long line surveys,

2. Converted to the form of the available proportion of the stock at each age in each year (for years where this is available),

Converted to the form of the available proportion of the stock at each length of males and of females treated separately. The convertion of estimates are made using the product of the matrix of length at age with the matrix of available stock structure at age.

Each of these sets of simulated estimates are compared to the equivalent observed outcomes of the long-line surveys and the parameter estimates are adjusted (using the solver tool in EXCEL) to produce the best fit possible. Goodness of fit between the estimates and the observed data are made on the basis of maximum likelihood.

Having obtained best fits to the parameters the model can then be used to estimate:-
1. The unfished level of spawning stock based upon average year-class

strength and a given maturity at age schedule and the levels of exploitation that would give 30%, 35% and 40% of the unfished spawning biomass.
2. The catch in the following few years to provide ABC?s and OFL levels according to the various management rules.

Consideration of the models structure and assumptions made in its formulation

The model used seems logically coherent and is in fact one of the few age based approaches possible given the sparse nature of available data from this stock. The spreadsheet implementation, while clearly provisional, has been constructed in a professional fashion and was as far as we could ascertain free from programing errors.

We considered whether more detailed and better known age based models (such as ad hoc tuned VPA?s) might be adapted to this stock but found that the data would not adequately support this approach.

The general class of catch at age based techniques tend to have the virtue that they predict the past absolute stock size (on the basis that when you have caught the lot you know how many there were!). They tend to converge towards these accurate past estimates at a rate dependent upon the level of mortality (i.e. how long it takes to catch the lot). Given the low mortality of sablefish this would argue for relatively slow convergence in the current model. Moreover, given that the current model only uses total commercial catch numbers data this property is not certain. However, we consider that the consequent fits to the age (or size) proportionate structure of the stock should secure this effect.

A virtue of the age based approach is that it avoids the need for having the estimate of total biomass in the sea (estimated by comparisons of long-line surveys and trawl swept area estimates of abundance) that is needed by the delay difference model previously used. A problem is that current estimates of stock size depend on available mortality or stock trend data. Hence the estimates of current stock size will largely depend mostly on the last few values of the survey or other trend data available (in this case the results of the long-line survey).

In the details of the model we note that an approximate form of the Baranov catch equation has been used but that this approximation is reasonable given 1) the low levels of mortality considered, 2) the distribution of fishing in the year. We also note that for those years for which available stock proportions at age are available it may be redundant to also fit the size data.

Given the sparse nature of the available sablefish data the model necessarily has to make a number of assumptions. These assumptions are needed to allow anage based model to be used. The assumptions are:-

1. The annual commercial catch numbers data are known exactly.
2. The commercial catch and the long-line survey both have the same age and size structure in a year.
3. The males and females form the same proportion of the population of each age in each year.

4. Natural mortality is constant with age and time.
5. That catchability at age forms a smooth function of a certain functional form
6. That the degree to which faith can be put in the various data sets is known.

Given the lack of data it is not possible to properly test these assumptions.
We note however that with respect to them:-
1. It is possible that catch was possibly under registered in years prior to the development of a discard observer program and that losses due to killer whales, long-line drop-offs etc. might be a reasonably constant source of additional unrecorded catch related mortality. This is thus one area where the details of the current model might be probed.
2. Commercial fishermen might concentrate on particular ages or sizes of fish while the survey attempts to sample all ages equally. Thus, differences might occur between the survey and the commercial size and age distributions. While it is not possible to test this assumption at present, it should certainly be addressed in the future with suitable sampling of commercial catches.
3. This problem is inherent in the current form of the data. If sex based differences in mortality or availability seem likely to be important it would be better to work with the proportion of each sex at each age or size. Such data are presumable available in the archives of the survey results. In the mean time, lacking this data, it is difficult to address this problem.
4. The assumption of constant natural mortality with age and year is a common feature of many fisheries assessment models. It has to be taken as a reasonable working assumption until more detailed predation based models (and their attendant data) can be developed.
5. This seems a reasonable simplifying assumption.
6. The degree of faith or weight to put on each of the three fitted data sets is difficult to establish. Pragmatically, like Fujioka et al (1996) we prefer to investigate the effects of different weights. We further consider that, given the differences between the age based model and the delay difference model, it would be worth fitting the age based model to the RPW series rather than to the RPN series since this might help generate some coherence between the approaches..

Hence, given the nature of the data we accept the need to make assumptions 3-5. We suspect that assumption 2 may be important but as yet no data for the scale of this problem are available to us. We are thus not able to investigate it meaningfully. We therefore concentrate our investigations on the effect of the assumption 1) to do with under registered commercial catch and 6)the relative weighting of the three fitted data sets and the choice of survey series.

Investigations of the possible effects of assumptions 1 and 6 on the biomass estimates and the catch forecast for 1996 and preliminary and tentative estimates for 1997.

Various model runs were made using modification of the NMFS spreadsheet AGESTR~1.XLS to question assumptions 1 and 6 above.

Run 1
Run 1 was a basic rerun of the NMFS model starting from different initial conditions. The results while not exactly the same were similar for all reasonable purposes and are included as a baseline. Results from this run are shown at figure 4.1.1-4.1.3. Figure 4.1.1 shows the 1996 exploitable population together with the exploitation percentages and the corresponding catch estimates for the 30% adjust. (OFL), the 35% adjust. ABC and the preliminary (based upon 1996 population) 40% adjust new rule ABC. Results for the age based method (in black) are contrasted with the equivalent delay difference results (in white). Apart from rounding errors these results are essentially the same as those given in figure 1.

Figure 4.1.2 shows the equivalent results for 1997. Since these are not based upon the 1996 commercial catch and survey results. WE STRESS THAT THESE RESULTS (AND EQUIVALENT RESULTS FOR FURTHER RUNS) ARE ILLUSTRATIVE AND TENTATIVE and will certainly change as a result of the 1996 data being included in future analyses. We therefore do not recommend them as a basis for making commercial decisions.

Figure 4.1.3 compares the trend of the scaled available biomass estimated by the model to the Relative Population Weight (RPW) survey estimate of biomass. It will be noted that the model estimate of biomass is lower than the RPW figure in the three most recent years. This difference is the main cause of the discrepancy between the age based and the delay difference estimate of the preliminary 1997 ABC. The figure also gives the estimate of the catchability of the survey series (i.e. the ratio between the survey and the model estimate of available biomass at the time of the survey. This is 1.20 and suggests that the survey calibration to total abundance (via swept area estimates) may have resulted in over estimates of the RPW series (since generally we would not expect swept area catchabilities to exceed 1). Alternatively it might suggest that the model was underestimating the available population. Possible reasons for this might perhaps be that catch data were underestimates or that the wrong level of natural mortality was adopted.

The model parameters were fitted to each of the data sets with each set given an equal weighting factor The goodness of fit measures were 0.165 for the RPN index, 0.268 for the proportion at age data and 1.367 for the proportion at size data. The total goodness of fit measure which the parameters are chosen to minimize (1.800= 0.165 + 0.268 + 1.367) is dominated by the proportion at size data. It thus seem worth either down weighting this or up weighting the survey index to see if this will result in a closer fit to the survey index particularly in the more recent years. This forms the basis of the next run.

Run 2
Run 2 is as run 1 except that the proportion at size data is given a weighting of 0.1 in order to set it to a contribution to the total that is more in balance with the other two data sets.
Results equivalent to those for run 1 are shown in figures 4.2.1 to 4.2.3. In particular this modification results in the following changes relative to run 1.

1. Exploitable biomass estimates are higher in both 1996 and 1997.

2. Exploitable percentages at the 30%,35% and 40% adjust levels are all estimated as larger in both 1996 and 1997.
3. Catch estimates are larger and more comparable to the DD model.
4. Biomass trend in recent years are marginally less steep and catchability marginally lower at 1.19.

The goodness of fit measures were 0.136 (was 0.165) for the RPN index, 0.146 (was 0.268) for the proportion at age data and 1.784 (was 1.367) for the proportion at size data. The total goodness of fit 0.460 measure = 0.136 +0.146 +1.784*0.1 which the parameters are chosen to minimize is now fairly evenly balanced between the results from the three data sets and has allowed the model to adjust better to the RPN and age data at the expense of the fit of the proportion at size data. This may be better than run 1 in that we would prefer to have catch at age data given more weight than catch at length data where both exist.

Run 3
Run 3 is as run 1 except that the commercial catch weight has been arbitrarily increased by 10% in all years, to see the effect this has on the model results. This is an attempt to see what effects under registering of catches (due to unrecorded catch or discards or non catch fishing deaths such as drop outs or killer whale interference) might have. Results equivalent to those for run 1 are shown in figures 4.3.1 to 4.3.3. In particular this modification results in the following changes relative to run 1.
1. Exploitable biomass estimates are higher in both 1996 and 1997.
2. Exploitable percentages at the 30%,35% and 40% adjust levels are all estimated as similar to those of run 1 in both 1996 and 1997.
3. Catch estimates are larger and more comparable to the DD model.
4. Biomass trend in recent years are similar to run 1 and catchability is lower at 1.09 by about 10% compared to run 1.

The goodness of fit measures were 0.165 (was 0.165) for the RPN index, 0.268 (was 0.268) for the proportion at age data and 1.367 (was 1.367) for the proportion at size data. The total goodness of fit 1.800 measure is thus the same as in run 1. These results indicate that increasing the catch by 10% in all years has scaled the model up to higher biomasses and hence catches but otherwise has left trends unchanged.

Run 4
Run 4 is as run 1 except that the commercial catch weight has been arbitrarily increased each year from 1979 to 1990 by 10% to see the effect this has on the model. Commercial catch weight was however left as recorded for years 1991-1995. This is an attempt to see what effects the under registering of catches, due to the unrecorded discards of earlier years, might have. Results equivalent to those for run 1 are shown in figures 4.4.1 to 4.4.3. In particular this modification results in the following changes relative to run 1.
1. Exploitable biomass estimates are slightly higher in both 1996 and 1997 than run 1.
2. Exploitable percentages at the 30%,35% and 40% adjust levels are all

estimated as similar to those of run 1 in both 1996 and 1997.
3. Catch estimates are slightly larger than run 1.
4. Biomass trend in recent years are similar to run 1 and catchability is lower at 1.16 compared to run 1.

The goodness of fit measures were 0.157 (was 0.165) for the RPN index, 0.271 (was 0.268) for the proportion at age data and 1.359 (was 1.367) for the proportion at size data. The total goodness of fit 1.787 measure is thus marginally smaller than that of run 1. These results indicate that increasing the catch by 10% in earlier years has caused relatively few changes to the model.

In the previous runs it is noticeable that the scaled estimates of available population are lower than the RPW index in recent years. Generally we would expected a model fitted to trend data such as the RPW to approximate to the index very closely in the latest years. This is because the final years population can be increased or decreased without violating the fit to the catch data. These populations thus tend to be determined by the population trend data. In the previous fits the population trend data was given by the RPN index. This suggests that using the RPW index in the fitting may reconcile the model to this trend index. Since this index was used in the past with the DD model, adopting it here for the AB model may increase comparability between the models.

Run 5
Run 5 is as run 1 except that the model is fit to RPW data rather than the RPN data.
Results equivalent to those for run 1 are shown in figures 4.5.1 to 4.5.3. In particular this change results in the following changes relative to run 1.
1. Exploitable biomass estimates are higher than run 1 in both 1996 and 1997 and are very comparable to the DD results.
2. Exploitable percentages at the 30%,35% and 40% adjust levels are all estimated as marginally larger than run 1 in both 1996 and 1997.
3. Catch estimates are larger than run 1 and more comparable to, though larger than, the DD model.
4. Biomass trend in recent years is less steep than run 1 and accords well with the RPW trend. Catchability is lower at 1.0 which means the survey and model estimates almost coincide.

The goodness of fit measures were 0.151 (was 0.165) for the RPW index, 0.265 (was 0.268) for the proportion at age data and 1.381 (was 1.367) for the proportion at size data. The total goodness of fit 1.797 measure which the parameters are chosen to minimize is just smaller but broadly comparable to that for run 1. Thus using RPW data has not adversely affected the overall fit and has marginally improved the trend data fit. Thus the RPW data seems as valid as the RPN data for model fitting and may be better in the sense that they have been used in past models and may therefore increase comparability.

Run 6

Run 6 is as run 1 except that the RPW index is used instead of the RPN index and the proportion at size data is given a weighting of 0.1 in order to set it to a contribution to the total that is more in balance with the other data.
Results equivalent to those for run 1 are shown in figures 4.6.1 to 4.6.3. In particular this change results in the following changes relative to run 1.

1. Exploitable biomass estimates are much higher in both 1996 and 1997 than run 1 and higher than the DD model..
2. Exploitable percentages at the 30%,35% and 40% adjust levels are all estimated as higher than in run 1 or the DD model in both 1996 and 1997.
3. Catch estimates are much larger than run 1 and higher than the DD model.
4. Biomass trend in recent years are marginally less steep. The biomass trend in at the 1986 peak was lower and the catchability marginally lower than run 5 at 0.96.

The goodness of fit measures were 0.134 (was 0.165) for the RPW index, 0.138 (was 0.268) for the proportion at age data and 1.755 (was 1.367) for the proportion at size data. The total goodness of fit measure 0.448= 0.134 +0.138 +1.755 *0.1 which the parameters are chosen to minimize is now fairly evenly balanced between the results from the three data sets and has allowed the model to adjust better to the RPW and age data at the expense of the size data. As with run 2, this may be better in that we would prefer to have catch at age data given more weight than catch at length data where both exist.

Summary of Runs
Modifications considered to the basic run shown at run 1 consider changing the weighting of the size data Runs 2 and 6, Changing the catch data, runs 3 and 4, changing the trend data from the RPN index to the RPW index. runs 5 and 6. As might have been anticipated, changes in each of these factors increased exploitable biomass and hence catch estimates and decreased the estimate of catchability of the RPW index. As might also have been anticipated, these factors reinforce one another when they are used in combination.
Reasoned arguments might be made for adopting all three changes to the age based assessment.

The relative proportions of stock at size data might be down-weighted because it appears to be the most variable, because it will be most affected by the sex ratio at age assumption and because it in some sense duplicates the more valuable relative proportions of stock at age data. However the proper degree of down-weighting would itself be open to argument.

The commercial catch weight might be plausibly increased to account for under registered landings. However a fixed scaling of catch mostly serves to scale the stock by an equivalent amount. Thus if discards were, for example, a constant proportion of the catch then including them would increase the stock and future catch estimates but leave estimates of the predicted landed catch unchanged unless the practice of discarding could be eliminated.

The RPW index might be adopted as the trend data rather than RPN data because its use would make the model more consistent with the previously used delay difference model and because it relates more closely to catch estimates than does the RPN.

There seems therefore to be reasonable arguments why at least a partial inclusion of the first factor and the third factor might be adopted in future runs of the age based method. Conversely it is of course true that a diligent search might find plausible factors that wouldt decrease the catch predictions. A likely candidate would be a decrease in the natural mortality estimate used. It is clear, however, that results from the age based model can alter as a result of changes in the choice of data sets and the relative weighting they are given in the fit.

It is also worth noting the similarities between the various runs as well as the differences. All agree on the broad size of the exploitable biomass (from 161,000mt to 225,000mt. All agree that current exploitation levels are at about the 10% level and all confirm the general trends in biomass which increased to a peak in 1986-1987 has been decreasing since. The catch predictions for 1997 seem on present data to be more pessimistic than those for 1996 but even more difference in catch prediction is generated by the shift from the old 35% ABC rule to the new 40% ABC rule than by biomass change. There are thus some broad truths that can be utilized to construct sound management policies for the sablefish stock.

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From J.G.POPE@dfr.maff.gov.uk Fri Nov 8 12:16:10 1996
Return-Path: J.G.POPE@dfr.maff.gov.uk
Received: from gatekeeper.maff.gov.uk (gatekeeper.maff.gov.uk [148.252.1.1]) by emin28.mail.aol.com (8.6.12/8.6.12) with ESMTP id MAA08669 for <NRCseattle@AOL.COM>; Fri, 8 Nov 1996 12:16:03 -0500
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From: John Pope FSMG DFR <J.G.POPE@dfr.maff.gov.uk>
Message-ID:
<0007716F.MAI*/I=JG/S=POPE/OU=DFR/O=MAFF/PRMD=MAFF400/ADMD=ATTmail/C=GB/@MHS>
To: Lee Alverson <NRCseattle@AOL.COM>
Subject: Sablefish 4

5 Summary and Conclusions.

The preliminary estimate of the 1997 ABC is smaller than the 1996 ABC. This is partly because the biomass is predicted at a lower level by the new age based assessment method, but mainly because new rules for setting ABC?s have been adopted which are more precautionary than those used to set the 1995 ABC. It should be noted that the preliminary 1997 ABC was set on the basis of the population in 1996 and thus may change when the assessment is updated in the light of the 1996 survey results and the 1996 commercial catch. Extrapolation from existing trends would suggest that the final estimate may be lower than the provisional estimate.

Examination of the calculations made to calculate the survey biomass trend and the new age based assessment method did not reveal any errors. Rather we conclude they were made in a professional, albeit rather preliminary, fashion. We believe it might be worth while recalibrating the survey indices from first principles but do not expect that this would show major differences in trend.

Consideration of the new management rules for setting FOFL and FABC (and hence the basis of calculating OFL and ABC catches), suggests that it would be possible to argue that, with the new age based assessment method, the sablefish might qualify for consideration under the second tier of the rules. This could lead to a more generous interpretations of the OFL and ABC than were made in the preliminary assessment (calculated under the third tier rules).

Reworking the age based assessment with different input assumptions made it possible to reconcile the model predictions more closely with the RPW longline survey index of total stock abundance. In particular, giving less weight to the length distribution data and tuning the results to the RPW, rather than the RPN, increased the compatibility of the new method to the previous approach and resulted in modest but significant increases in ABC and OFL catch levels for 1997.

While a range of interpretations of the age based method exist, all broadly agree on the following points. These are:-

1. On average the RPW index is quite close to the estimated total biomass.

2. Biomass has declined in recent years from the high levels of the mid 1980?s

3. The current level of exploitation is comparatively low at about 10%

4. Current levels of spawning biomass are close to the calculated B30% level.

5. The stock structure is an ageing one due to the low recruitment of young fish in recent years.

The situation of the sablefish stock would therefore seem to be in control but the stock decline is of concern. At present it requires a careful rather than an excessively conservative management approach. Should the biomass continue to decline over the next few years with continuing low recruitments then more stringent measures will be required. The new management rules applied with discretion would seem adequate to this task. The fishing industry should not necessarily oppose reductions in fishing intensity, even though it requires short term losses, if it brings a promise of higher and more consistent yield over time and adds to the value of their IFQ certificates. However, they can reasonably request investigations to establish that the new management will achieve this result. They can also reasonably press for the assessment method and data to be improved to a standard where more optimal exploitation of the sablefish stock can be permitted under the new rules.

Hope you got all this John

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Received: from unknown(148.252.3.25) by gatekeeper.maff.gov.uk via smap (g3.0.3)
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Date: Sat, 9 Nov 1996 01:13:04 +0000
X400-Originator: J.G.POPE@dfr.maff.gov.uk
X400-Recipients: NRCseattle@AOL.COM
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From: John Pope FSMG DFR <J.G.POPE@dfr.maff.gov.uk>
Message-ID:
<00077170.MAI*/I=JG/S=POPE/OU=DFR/O=MAFF/PRMD=MAFF400/ADMD=ATTmail/C=GB/@MHS>
To: Lee Alverson <NRCseattle@AOL.COM>
Subject: Sablefish 5